TOPIC 07 · MAGNETIC AND THERMAL PROPERTIES

Magnetic and Thermal Properties

The origin and types of magnetism, domains and hysteresis, superconductivity, and the thermal properties: heat capacity, thermal expansion, thermal conductivity, thermal stress and thermal shock.

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Central idea: every magnetic property comes from the magnetic moments of electrons, and the type of magnetism depends on whether those moments exist in the atom, whether they are permanent, and how neighbouring moments are arranged: induced and opposite (diamagnetic), permanent and random (paramagnetic), parallel (ferromagnetic), antiparallel and equal (antiferromagnetic) or antiparallel and unequal (ferrimagnetic). In the ferromagnetic and ferrimagnetic materials the moments group into domains, and the motion of the domain walls sets the shape of the hysteresis loop, so a material is a soft magnet or a hard one. The thermal properties follow from what heat does to the lattice: it raises the vibrational energy (heat capacity), it increases the mean interatomic distance (thermal expansion), it is carried by electrons and phonons (thermal conductivity), and when expansion is restrained or uneven it creates stresses (thermal stress and thermal shock).

This page follows the single official session of Topic 7 (Session 15) slide by slide. Part I covers the magnetic properties (Slides 1 to 32) and Part II the thermal properties (Slides 33 to 48). Every section names the slides it comes from. Figures that appear on the slides are copied from them as PNG crops, and each caption names its slide; the two drawings that are not on the slides (the asymmetric potential well and the thermal-shock stress profile) are extensions and their captions say so. Anything that is not printed on the slides is marked as an extension. The photographs of the slides (uses of magnets, levitation, trains, thermocouples) are not reproduced. Problems 11 and 12 of the sheet `Problems T6andT7` are the magnetic problems of this topic, and they are solved on the problems page of this topic.

Part I — Magnetic Properties (Session 15, Slides 1–32)

§ 1 Uses and overview [Session 15 Slides 1-2]

Slide 1 is the title and the agenda of the magnetic part: introduction, basic concepts, types of magnetism, magnetic domains, hysteresis cycles, and superconductivity and magnetism. Slide 2 lists the uses of magnetic materials with pictures: the compass, ferrofluids, permanent magnets, and the magnetic coverings used in stealth technologies.

§ 2 Origin of the magnetic behaviour [Session 15 Slide 3]

Magnetism is the phenomenon by which materials induce forces of attraction or repulsion over other materials, and it has its origin in the movement of electrically charged particles [Session 15 Slide 3]. The macroscopic magnetic properties of a material are a consequence of the magnetic moments associated with its individual electrons. Each electron has two contributions:

- **Orbital magnetic moment** $= m_l \, \mu_B$, from its motion around the nucleus.
- **Spin magnetic moment** $= \pm \mu_B$, from its spin (two possible directions).

Here m_l is the magnetic quantum number of the electron (the slide writes "magnetic quantum moment") and μ_B is the **Bohr magneton**:

$$\mu_B = 9.27 \times 10^{-24} \ \mathrm{A \cdot m^2}$$

The magnetic moment of an atom is the sum of the magnetic moments of all its electrons. If the net moment of the atom is not null, the atom behaves like a **magnetic dipole**. The slide tabulates the moment of the 3d transition elements from the electronic structure of their 3d shell, which counts its unpaired spins (the 4s electrons are not counted, and the moment of an atom is not the moment per atom of the solid metal: compare Fe, Co and Ni here with the values of Section 4.3):

Element	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu
Atomic number	21	22	23	24	25	26	27	28	29
Moment in μ_B	1	2	3	5	5	4	3	2	0

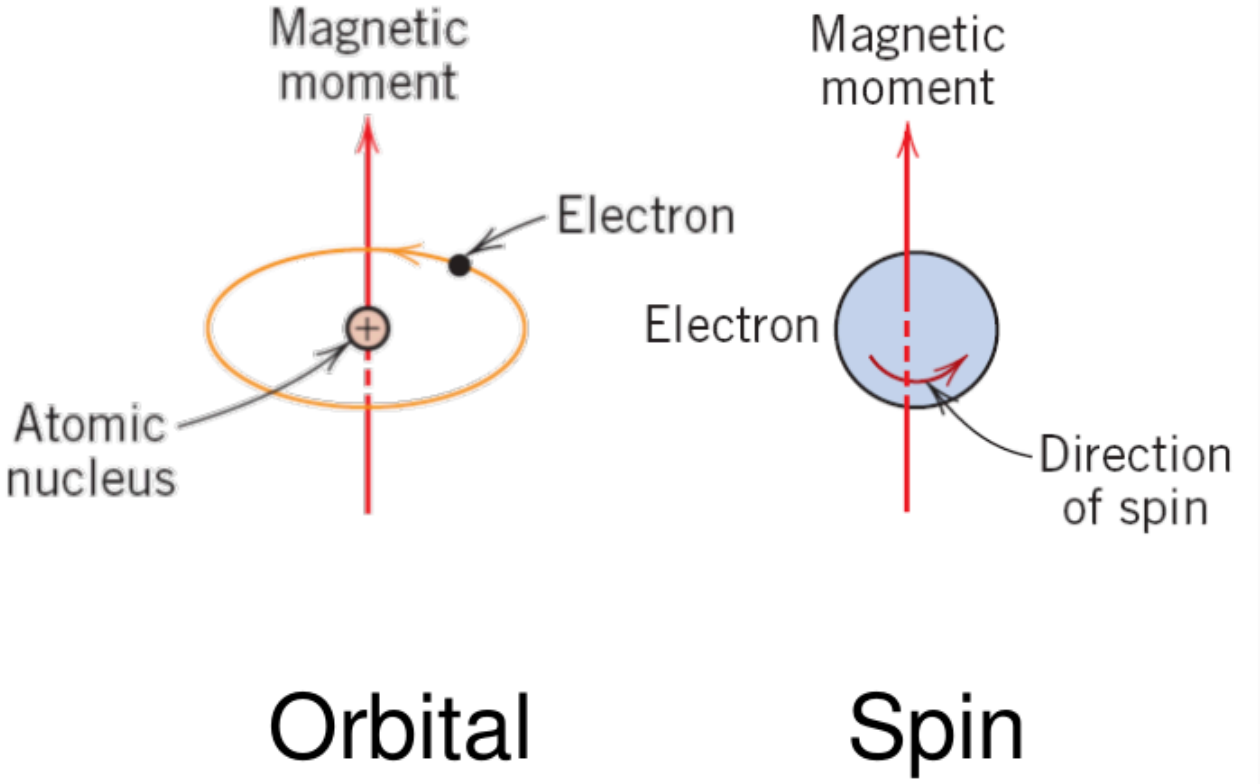


Figure 1. Origin of the magnetic moment of an atom: the orbital moment of the electron moving around the nucleus and the spin moment of the electron spinning about its axis, copied from Session 15 Slide 3.

DEFINITION OF THE BOHR MAGNETON

The slide gives only the value of μ_B. The definition $\mu_B = e\hbar/2m_e$ is an extension, and it reproduces the value above.

§ 3 Basic concepts: field, induction, magnetization [Session 15 Slide 4]

Slide 4 defines the quantities of the magnetic response of a material in SI units. Some later slides also print other units: gauss (Slide 23), emu/cm³ and Oe (Slide 24), cal (Slides 35 and 41) and Btu (Slide 43).

- H is the **magnetic field strength**, in A/m.
- B is the **magnetic induction**, also called magnetic flux density (the slides also write "inductance"), in Wb/m² or tesla.
- μ is the **permeability** of the medium through which H passes and in which B is measured, in H/m (or Wb/(A·m)).
- M is the **magnetization** (the magnetic moment per unit volume), in A/m.
- χ_m is the **magnetic susceptibility** (dimensionless).

$$B = \mu H \qquad \text{and in vacuum} \qquad B = \mu_0 H \qquad \text{with} \quad \mu_0 = 1.257 \times 10^{-6}\ \mathrm{H/m}$$

$$\mu_r = \frac{\mu}{\mu_0} \qquad B = \mu H = \mu_0 H + \mu_0 M \qquad M = \chi_m H$$

The relative permeability μ_r is the ratio of the permeability of the material to that of vacuum. Substituting $M = \chi_m H$ in $B = \mu_0 H + \mu_0 M$ gives $B = \mu_0(1 + \chi_m)H$, and comparing with $B = \mu_r \mu_0 H$,

$$\mu_r = 1 + \chi_m \qquad \Rightarrow \qquad \chi_m = \mu_r - 1$$

(derived here from the two printed relations). The figure of the slide plots the induction against the field: in vacuum $B = \mu_0 H$, and in a material the line differs by the term $\mu_0 M$, which is the contribution of the material (higher for $M > 0$, lower for the diamagnetic $M < 0$).

DIMENSIONAL CHECK

$[\mu_0 H] = (\mathrm{H/m})(\mathrm{A/m}) = \mathrm{V \cdot s/m^2} = \mathrm{T}$, the unit of B. χ_m and μ_r have no units, because M and H are both in A/m.

§ 4 Types of magnetism [Session 15 Slides 5-13]

Slide 5 gives the overview. Materials are grouped by the value of μ_r: the **ferromagnetic** and **ferrimagnetic** ones have $\mu_r \gg 1$ and $\chi \approx 10^5$ to 10^6, while the **paramagnetic** and **diamagnetic** ones have μ_r close to 1 and a susceptibility of about 10^{-5} in absolute value. The **antiferromagnetic** materials are non-magnetizable. The chart of the slide plots the induction against the applied field with the line of vacuum ($\chi = 0$) as reference [Session 15 Slide 5]:

- (1) Diamagnetic ($\chi \sim -10^{-5}$): Al_2O_3, Cu, Au, Si, Ag, Zn. The line lies below that of vacuum.
- (2) Paramagnetic: Al, Cr, Mo, Na, Ti, Zr. The line lies slightly above that of vacuum.
- (3) Ferromagnetic and ferrimagnetic (χ as large as 10^6): Fe_3O_4, $NiFe_2O_4$, ferrite (α), Co, Ni, Gd. The line is far above, with $B \cong \mu_0 M$ for $H \ll M$.

PRINTED SIGN OF THE DIAMAGNETIC SUSCEPTIBILITY

The text of Slide 5 writes the diamagnetic susceptibility as $\chi \approx 10^{-5}$ without the sign, while the chart on the same slide and Slide 6 give $\chi_m < 0$. The sign is negative.

CHROMIUM IN TWO LISTS

Cr is listed as paramagnetic on Slide 5 and as antiferromagnetic on Slide 10. Slide 23 resolves it: its Néel temperature is 310 K, so it is antiferromagnetic below 310 K and paramagnetic above. By the same table MnO (116 K) and α-Mn (100 K) are paramagnetic at room temperature (this reading is an extension).

4.1 Diamagnetism [Session 15 Slides 6-7]

In a diamagnetic material there are no dipoles in the absence of an external field. In the presence of a field, dipoles are induced that are aligned **opposite** to the field direction. The external field induces a change in the speed of the orbital movement of the electrons, so the effect is **not permanent**: it exists only while the field is applied. The induced moment is very small and opposite to the applied field [Session 15 Slide 6]:

$$\mu_r < 1 \qquad \chi_m < 0 \qquad B_{dia} \cong \mu_0 H \qquad B_{dia} < B_0$$

where $B_0 = \mu_0 H$ is the induction in vacuum.

Examples on the slide: gold, silver and copper.

Slide 7 is titled "The frog incident" and only prints the link of a video (listed at the end of Part I). The slide does not describe it. The title points to the well-known demonstration of the diamagnetic levitation of a living frog in a very strong magnetic field; this reading is outside the slide.

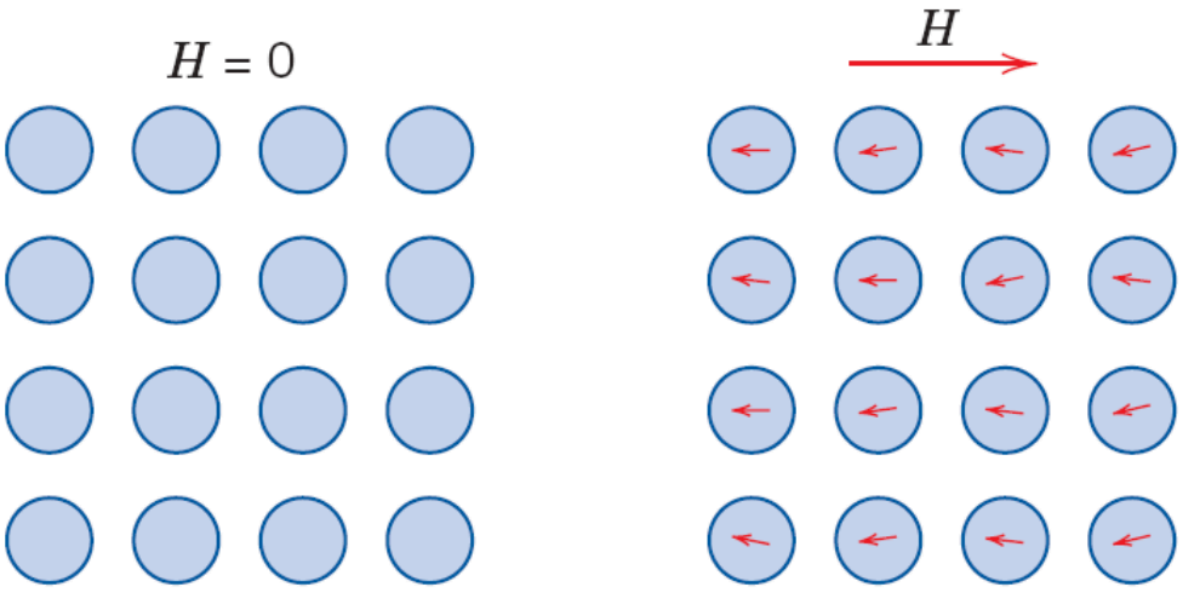


Figure 2. Diamagnetic material: no dipoles without a field (H = 0) and induced dipoles opposite to an applied field H, copied from Session 15 Slide 6.

4.2 Paramagnetism [Session 15 Slide 8]

A paramagnetic material has atoms with **permanent** magnetic dipoles that are **randomly oriented**, so the material has no permanent net magnetization. When an external field is applied the dipoles turn and align with its direction. The induced field is relatively small and its effect is **added** to the applied field [Session 15 Slide 8]:

$$\mu_r > 1 \qquad 10^{-5} \leq \chi_m < 10^{-2} \qquad B_{para} \cong \mu_0 H \qquad B_{para} > B_0$$

Examples on the slide: aluminium and titanium.

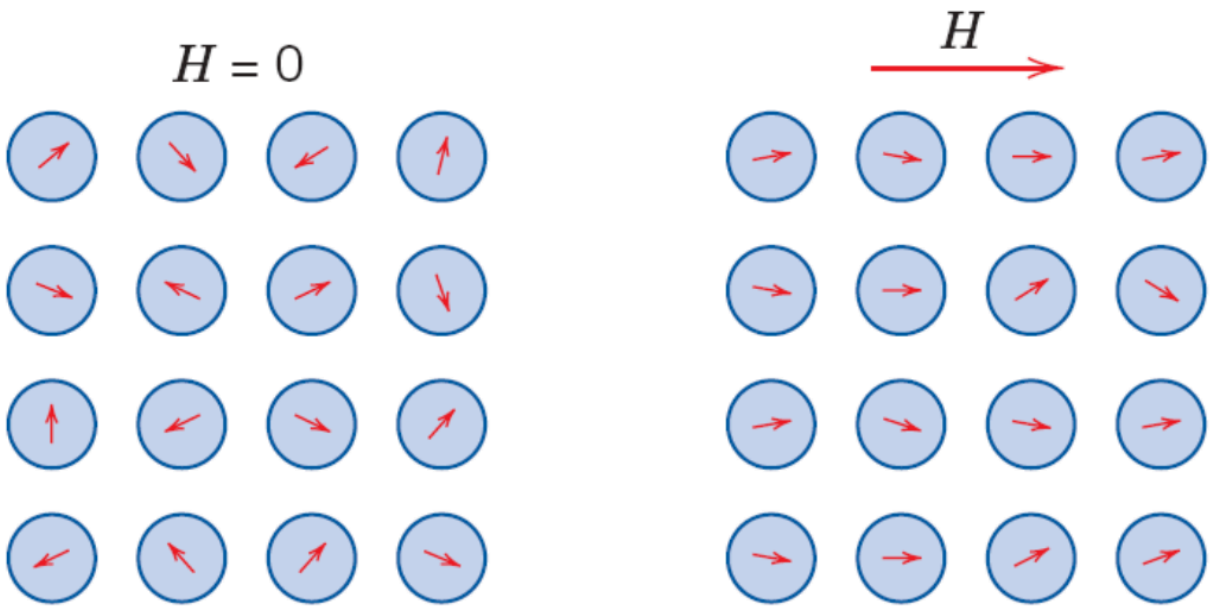


Figure 3. Paramagnetic material: permanent dipoles randomly oriented without a field (H = 0) and aligned with an applied field H, copied from Session 15 Slide 8.

4.3 Ferromagnetism [Session 15 Slide 9]

A ferromagnetic material has magnetic dipoles **permanently oriented** (parallel) in the absence of an external field, so it shows a large permanent magnetization (within each domain: Section 5 explains why a piece that was never magnetized has no net magnetization). The slide states that the permanent magnetic moment of ferromagnetic materials is due fundamentally to the **spin** magnetic moments, with a small contribution from the orbital moments [Session 15 Slide 9]. The susceptibility is very large, up to 10^6:

$$\chi_m \gg 1 \quad (\text{up to } 10^6) \qquad \Rightarrow \qquad M \gg H \qquad \Rightarrow \qquad B \cong \mu_0 M$$

The slide tabulates the magnetic moment at 0 K of the ferromagnetic metals:

Metal	Fe	Co	Ni	Gd
Magnetic moment in μ_B	2.22	1.72	0.60	7.12

The examples of the slide are iron, nickel and cobalt; gadolinium appears in this table and in the table of Curie temperatures (Section 9).

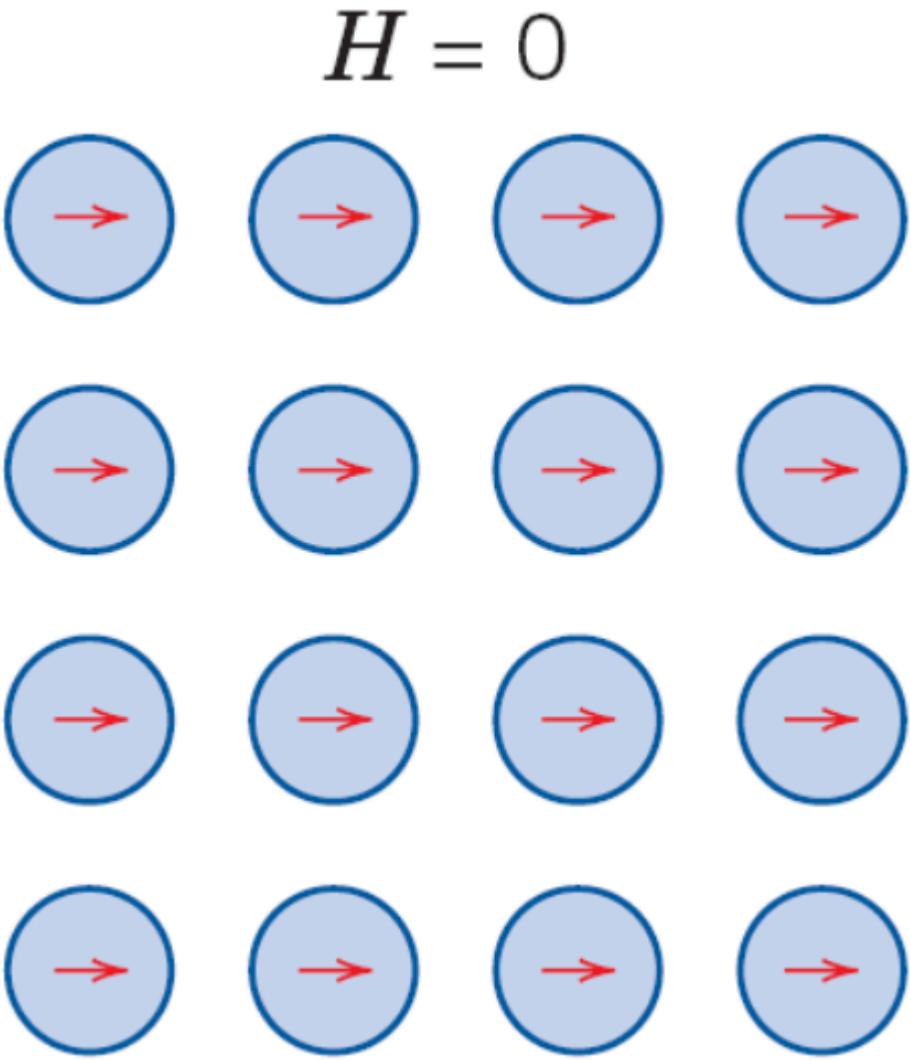


Figure 4. Ferromagnetic material: dipoles parallel to each other with no applied field (H = 0), copied from Session 15 Slide 9.

4.4 Antiferromagnetism [Session 15 Slide 10]

In an antiferromagnetic material the permanent magnetic dipoles are oriented in an **antiparallel** way, so the solid as a whole has no net magnetic moment. The examples are Cr, Mn and MnO. The slide shows the antiparallel alignment of the spin moments of the Mn^{2+} ions in manganese oxide, with the O^{2-} ions in between [Session 15 Slide 10].

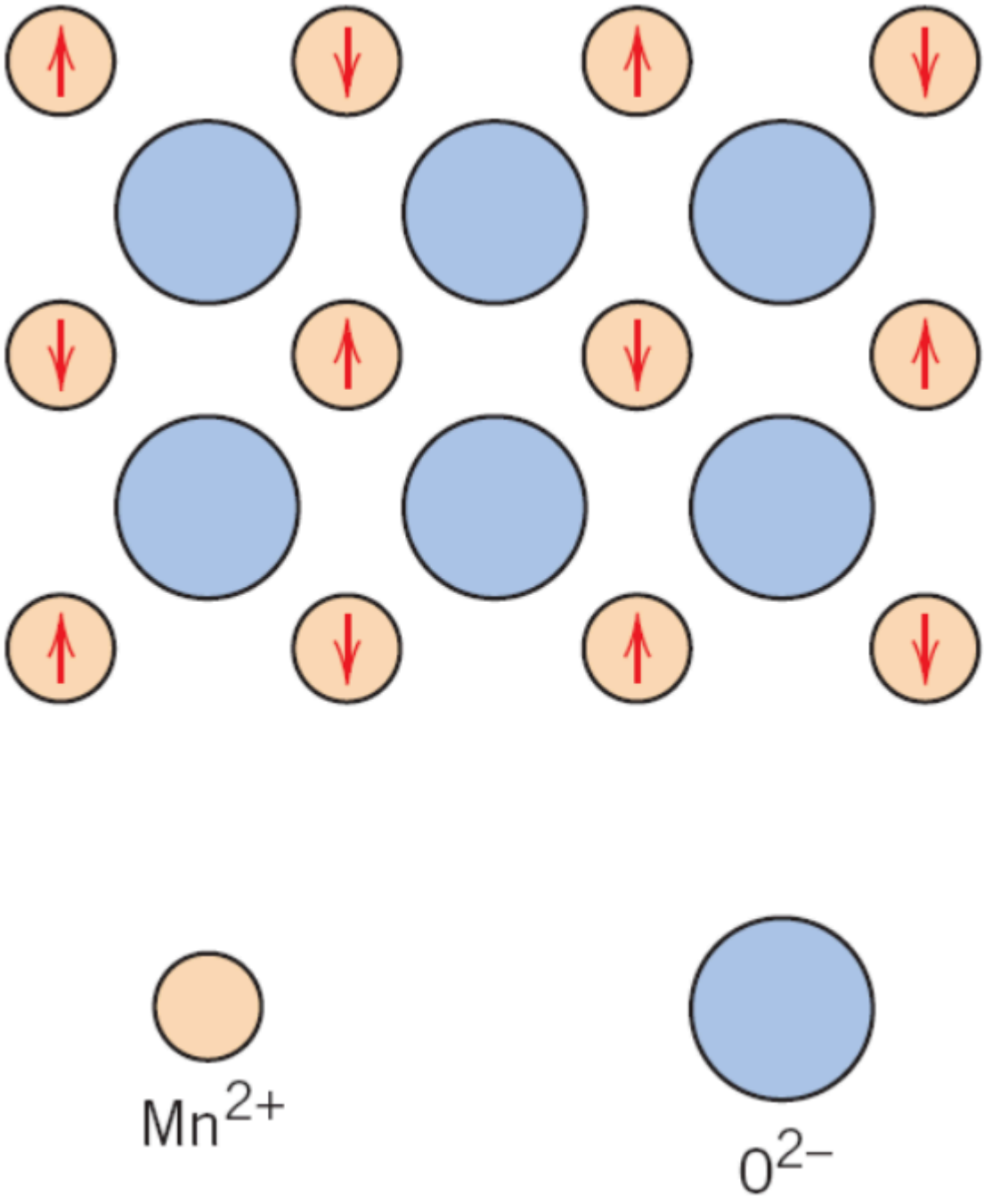


Figure 5. Antiparallel alignment of the spin moments of the Mn2+ ions in manganese oxide, with the O2- ions in between, copied from Session 15 Slide 10.

4.5 Ferrimagnetism [Session 15 Slides 11-12]

Ferrimagnetism is present in **ceramic** materials with permanent dipoles of **different intensity** aligned in an antiparallel way, so a net moment remains. The ferrimagnetic ceramics are the **ferrites** (compounds of formula AB_2O_4), the **garnets** ($M_3Fe_5O_{12}$) and the compounds $AB_{12}O_{19}$ [Session 15 Slide 11]. The slide shows the arrangement of the cations in the oxygen lattice (Fe^{2+} and Fe^{3+} in octahedral sites, Fe^{3+} in tetrahedral sites) and the net spin magnetic moment of each cation:

Cation	Fe^{3+}	Fe^{2+}	Mn^{2+}	Co^{2+}	Ni^{2+}	Cu^{2+}
Net spin moment in μ_B	5	4	5	3	2	1

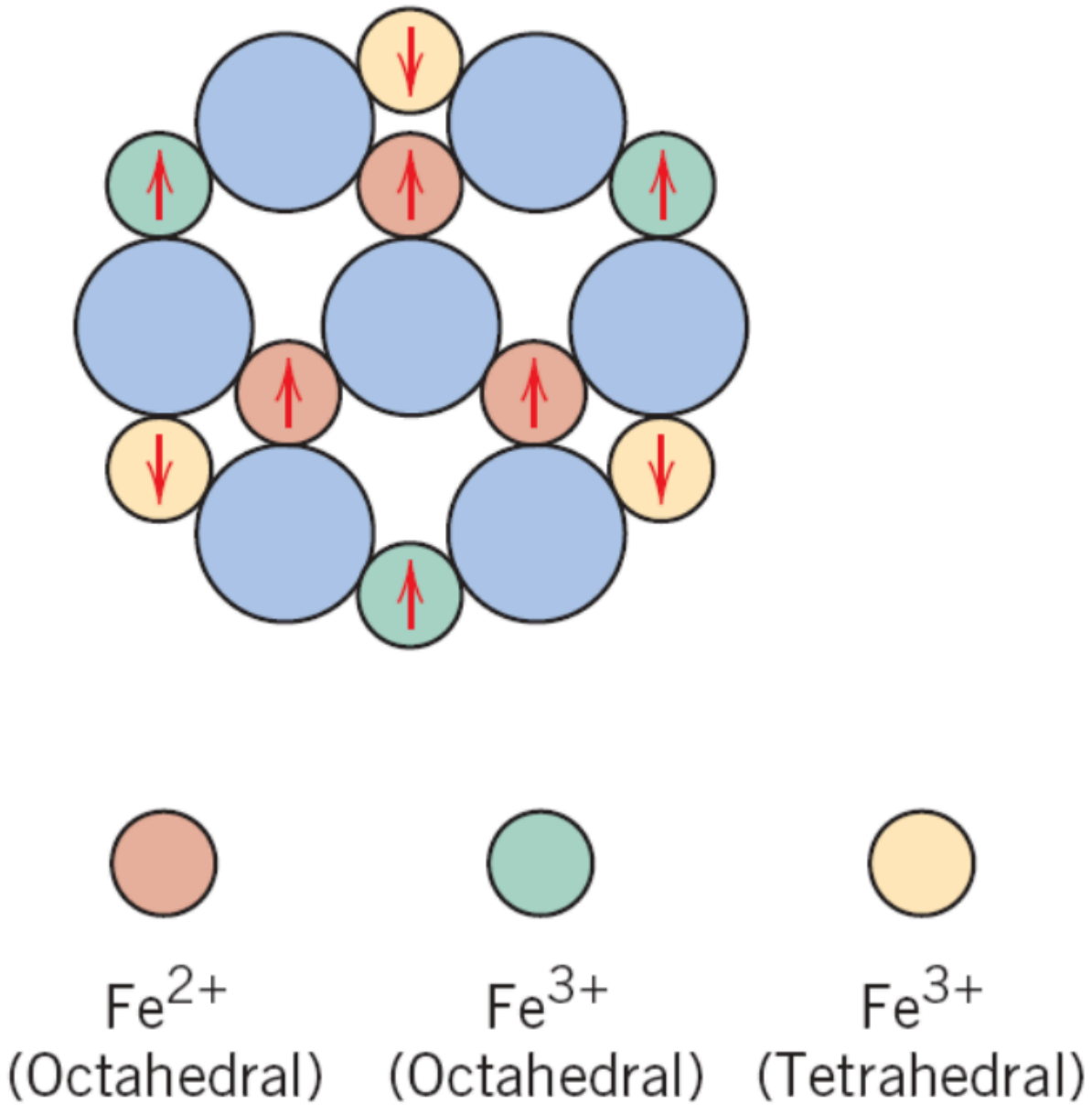


Figure 6. Ferrimagnetic ferrite: spin moments of the Fe2+ and Fe3+ ions in octahedral sites and of the Fe3+ ions in tetrahedral sites of the oxygen lattice, copied from Session 15 Slide 11.

Magnetite, Fe_3O_4 (extension, built from the table above). Magnetite is the ferrite $Fe^{3+}(Fe^{2+}Fe^{3+})O_4$: one Fe^{3+} ion in a tetrahedral site and one Fe^{2+} and one Fe^{3+} ion in octahedral sites per formula unit. The two Fe^{3+} moments are antiparallel and cancel, $5\,\mu_B - 5\,\mu_B = 0$, so the net moment per formula unit is that of the Fe^{2+} ion, $4\,\mu_B$.

Slide 12 lists commercial ferrite compositions ($MgFe_2O_4$, Mg-Zn, $MnFe_2O_4$, Mn-Fe, Mn-Zn, $NiFe_2O_4$ and the lithium ferrite) and commercial garnet compositions (yttrium iron garnet $Y_3Fe_5O_{12}$ and its aluminium, chromium, lanthanum and praseodymium variants), with the comments on which site each substituting ion prefers.

Name	Composition	Comments
Magnesium ferrite	$MgFe_2O_4$	
Magnesium–zinc ferrite	$Mg_xZn_{1-x}Fe_2O_4$	$0 < x < 1$.
Manganese ferrite	$MnFe_2O_4$	
Manganese–iron ferrite	$Mn_xFe_{3-x}O_4$	$0 < x < 3$.
Manganese–zinc ferrite	$Mn_xZn_{1-x}Fe_2O_4$	$0 < x < 1$.
Nickel ferrite	$NiFe_2O_4$	
Lithium ferrite	$Li_{0.5}Fe_{2.5}O_4$	Li^+ occurs in combination with Fe^{3+}, producing a $(Li_{0.5}Fe_{0.5})^{2+}Fe_2^{3+}O_4$ configuration.

SOME COMMERCIAL GARNET COMPOSITIONS

Name	Composition	Comments
Yttrium iron garnet (YIG)	$Y_3Fe_5O_{12}$	
Aluminum substituted YIG	$Y_3Al_xFe_{5-x}O_{12}$	Al^{3+} prefers tetrahedral sites.
Chromium substituted YIG	$Y_3Cr_xFe_{5-x}O_{12}$	Cr^{3+} prefers octahedral sites.
Lanthanum iron garnet (LaIG)	$La_3Fe_5O_{12}$	
Praseodymium iron garnet (PrIG)	$Pr_3Fe_5O_{12}$	Many of the pure garnets are not commercially prepared; all form at least limited solid solutions with each other (e.g., $Pr_xY_{3-x}Fe_5O_{12}$ with $x_{max} = 1.5$).

Figure 7. Some commercial ferrite and garnet compositions, copied from Session 15 Slide 12.

4.6 Summary of the types of magnetism [Session 15 Slide 13]

Slide 13 draws the dipoles of the five types and groups them by when they are oriented: ferromagnetic, antiferromagnetic and ferrimagnetic materials have **oriented dipoles in the absence of a magnetic field**; paramagnetic materials have dipoles **oriented only in the presence of a field**; diamagnetic materials have **no permanent dipoles, only induced dipoles**. Together with the previous slides:

Type	Dipoles	μ_r and χ_m	Examples
Diamagnetic	induced, opposite to H, not permanent	$\mu_r < 1$, $\chi_m < 0$ (about -10^{-5})	Au, Ag, Cu, Si, Zn, Al_2O_3
Paramagnetic	permanent, random, aligned by H	$\mu_r > 1$, $10^{-5} \leq \chi_m < 10^{-2}$	Al, Ti, Cr, Mo, Na, Zr
Ferromagnetic	permanent, parallel without a field	$\mu_r \gg 1$, χ_m up to 10^6	Fe, Co, Ni, Gd

Type	Dipoles	μ_r and χ_m	Examples
Antiferromagnetic	permanent, antiparallel and equal	no net moment, non-magnetizable	Cr, Mn, MnO
Ferrimagnetic	permanent, antiparallel and unequal	$\mu_r \gg 1$, χ_m about 10^5 to 10^6	ferrites, garnets, Fe_3O_4

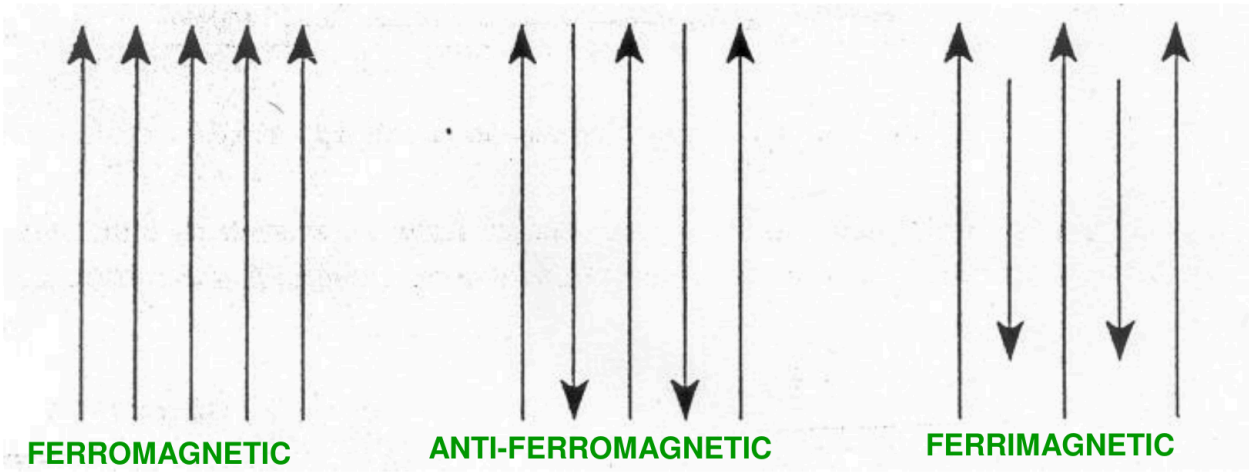


Figure 8. Dipoles of the ferromagnetic (parallel), antiferromagnetic (antiparallel and equal) and ferrimagnetic (antiparallel and unequal) arrangements, copied from Session 15 Slide 13.

§ 5 Magnetic domains and Bloch walls [Session 15 Slides 14–15]

A **magnetic domain** is a three-dimensional region of a ferromagnetic or ferrimagnetic material with uniform magnetization. Because of the domains, a ferromagnetic or ferrimagnetic material that has never been exposed to a magnetic field has a null net magnetization [Session 15 Slide 14].

A **Bloch wall** is a narrow transition region at the boundary between magnetic domains, over which the magnetization changes progressively from its value in one domain to its value in the next. The sizes given on the slide are:

Feature	Size
Domain	about 10 to 100 μm
Bloch wall	about 100 nm (approximately 300 atoms)

The domains are usually smaller than the grains: a small grain can be a single domain, and a larger grain contains several. The slide shows the domain pattern of an iron (001) single crystal film.

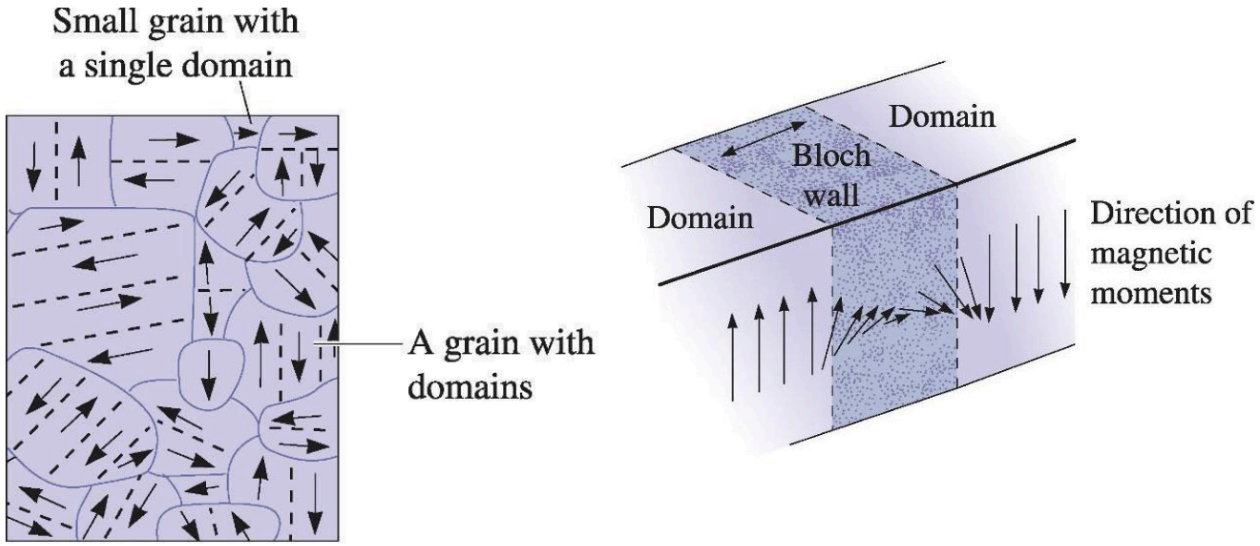
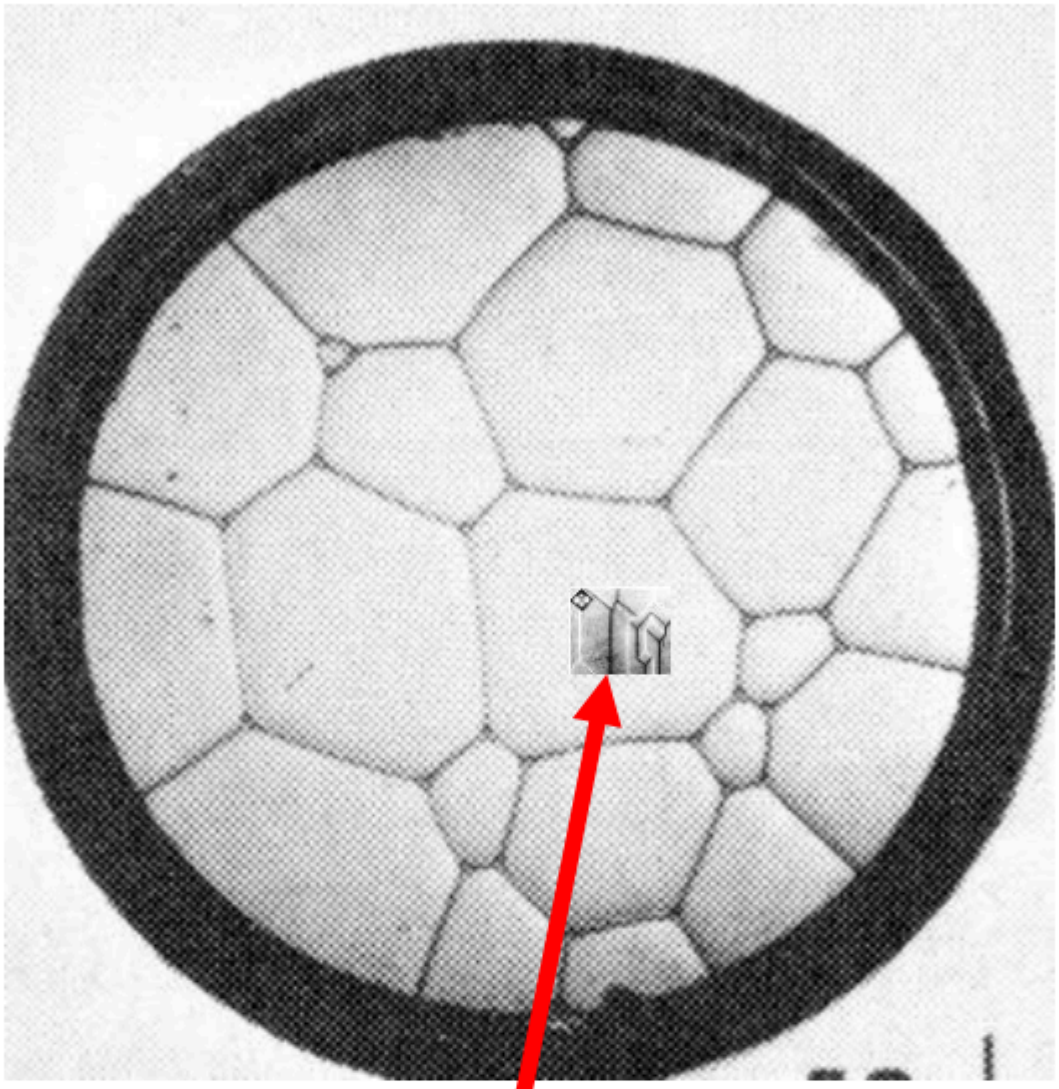


Figure 9. A grain with domains, a single-domain small grain and a Bloch wall between two domains, copied from Session 15 Slide 14.



Domains in iron

(001)

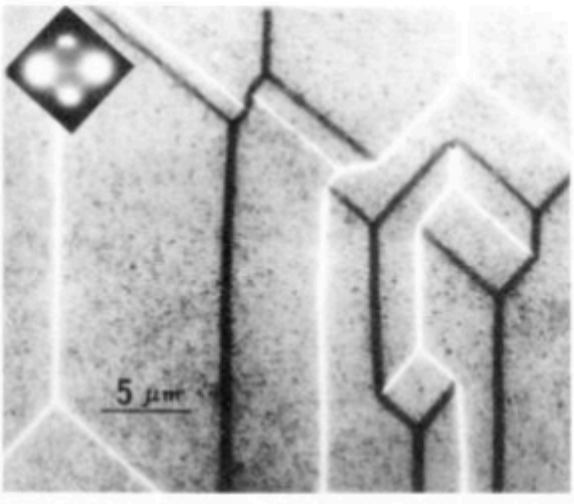


Fig. 15.16. Domain pattern observed in a (001) iron single crystal film of thickness 2500 Å, and



Figure 10. Domain pattern observed in a (001) iron single crystal film 2500 Å thick, copied from Session 15 Slide 14.

Slide 15 gives the reason for the domains: the magnetic domain structure is generated **spontaneously to decrease the magnetostatic energy** of the material. A single domain has a high magnetostatic energy (the stray field closes through the air around it); two domains have a lower one, four domains a lower one still, and the closure domains of the last panel have zero magnetostatic energy [Session 15 Slide 15].

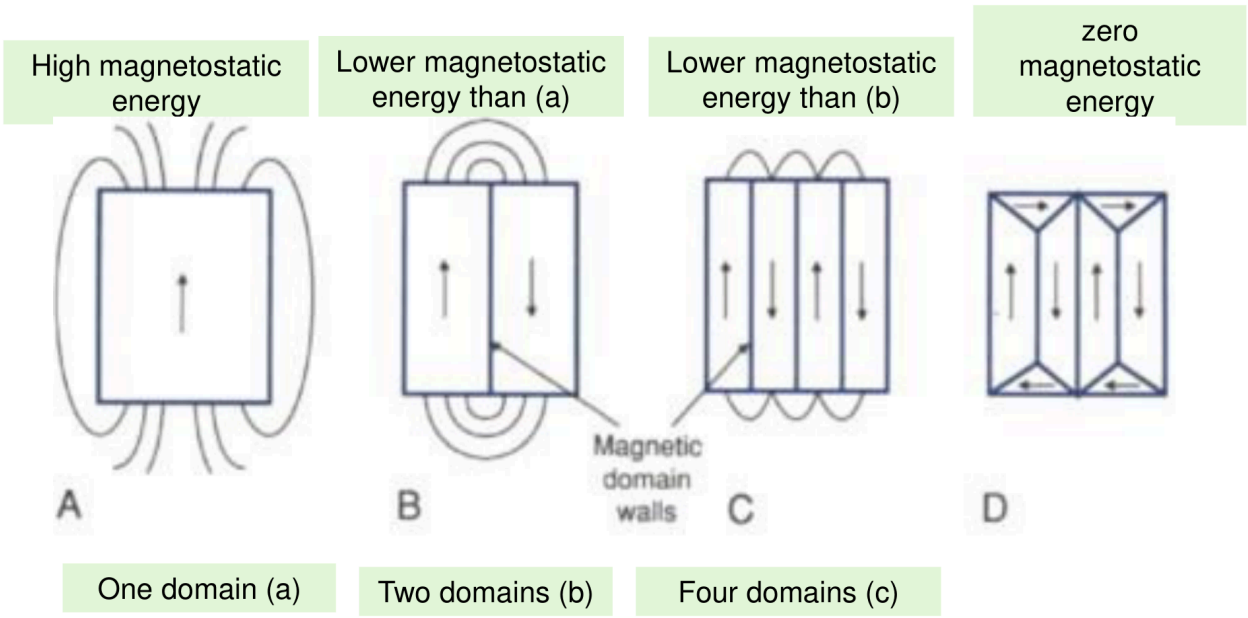


Figure 11. One domain (high magnetostatic energy), two domains, four domains and closure domains (zero magnetostatic energy), with the magnetic domain walls, copied from Session 15 Slide 15.

§ 6 Magnetization curve and permeability [Session 15 Slide 16]

When a magnetic field is applied to a magnetic material the magnetization increases slowly at first, then more rapidly as the domains begin to grow, and later it slows down because the domains must eventually **rotate** to reach saturation [Session 15 Slide 16]. The stages of the slide are:

1. **Random distribution of domains** (demagnetized state, $H = 0$).
2. **Growth of the favourable domains** and shrinkage of the unfavourable ones.
3. **More growth** of the favourable domains.
4. **Rotation of domains** towards the field direction.
5. **Saturation.**

The two permeabilities of the slide are the slopes of the curve: μ_i is the **initial permeability** (slope at the start) and μ_{max} is the **maximum permeability** (the line from the origin tangent to the curve, that is, the largest value of B/H); the permeability varies with H. The slide's axis reads "magnetization M or inductance B": strictly these slopes are permeabilities on the B curve. The **saturation magnetization** M_s is the magnetization that results when all the magnetic dipoles of the solid are aligned with the external field; the corresponding induction is the saturation flux density B_s.

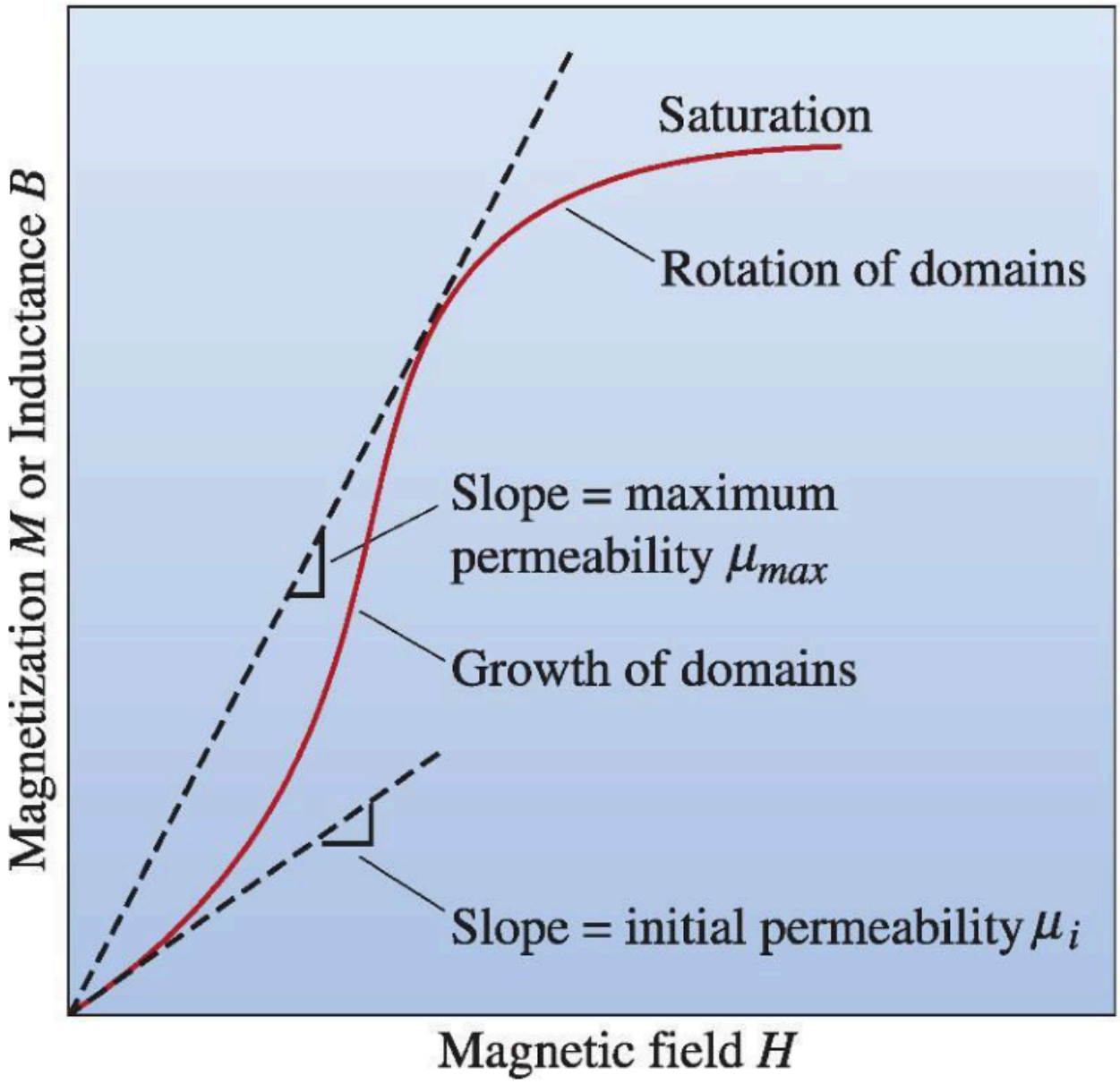


Figure 12. Magnetization M or induction B against the field H from the demagnetized state: initial permeability μi, maximum permeability μmax, growth of domains, rotation of domains and saturation, copied from Session 15 Slide 16.

§ 7 Hysteresis cycles and applications [Session 15 Slides 17-19]

7.1 The hysteresis loop [Session 15 Slide 17]

The slide draws the loop with the domains at each point. When the field is reduced from saturation, the magnetization does not follow the initial curve back, because the domains do not return to their random arrangement (the author's reading of the sketches). The loop of B against H defines two quantities:

- B_r is the **remanence**: the induction that remains when the field is brought back to zero.
- H_c is the **coercivity**: the reverse field needed to bring the induction to zero. It is a measure of the opposition to a change in orientation of the dipoles: the slide writes it as 1/(how easy it is to change the orientation of the dipoles).

7.2 Soft and hard magnetic materials [Session 15 Slide 18]

The materials are classified by the shape of their loop, with the limit $H_c = 10\,000$ A/m [Session 15 Slide 18]:

	Soft magnetic materials	**Hard magnetic materials**
Bloch walls	easy to move	difficult to move
Properties	high initial permeability, low H_c	high remanence, high B_s and H_c, low μ_i
Use	materials under alternating fields, to increase B and produce fast flux changes in coils and circuits with minimum energy losses	permanent magnets, because of their high resistance to demagnetization

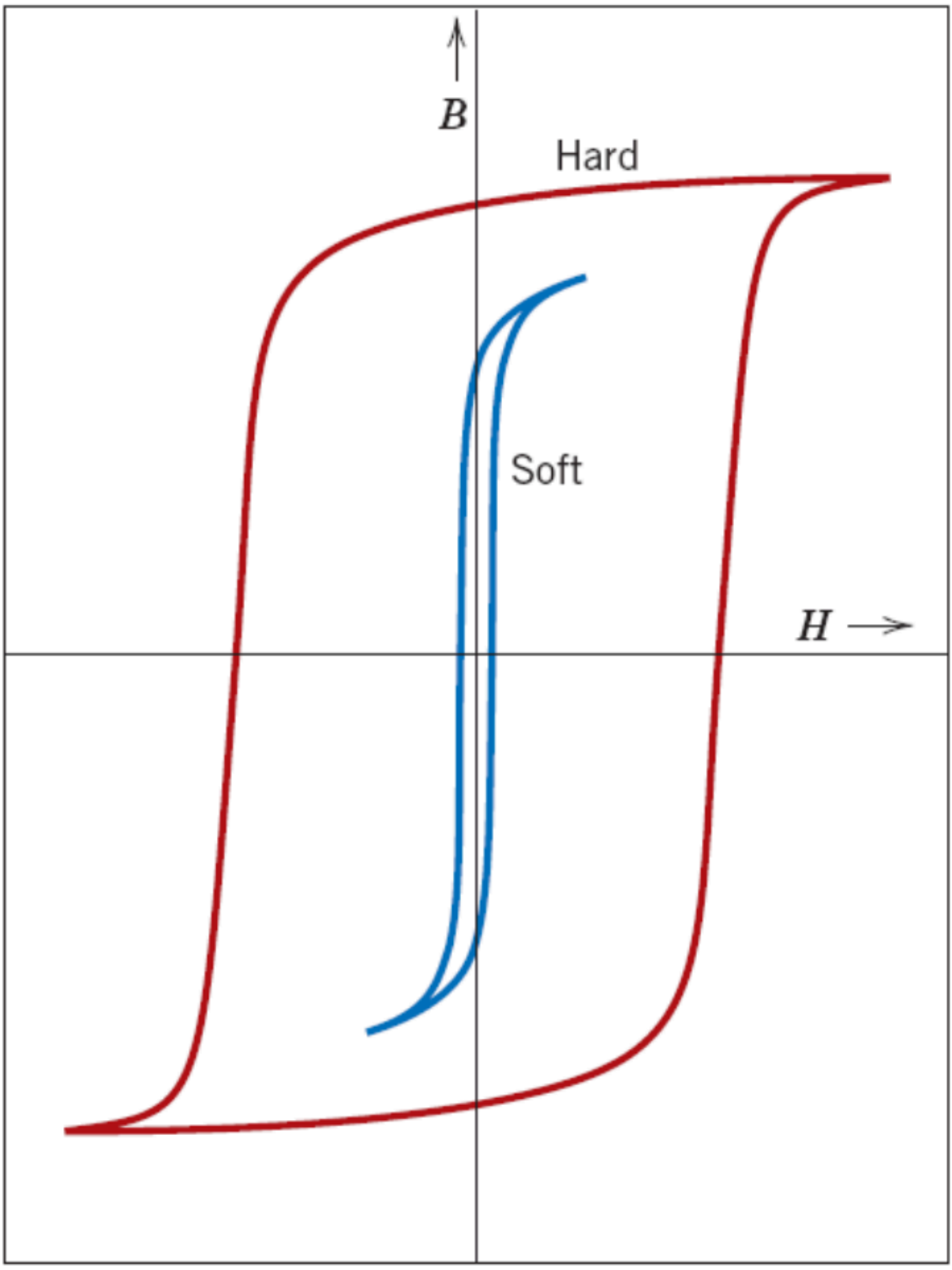


Figure 13. B-H hysteresis loops of a soft magnetic material (narrow loop) and a hard magnetic material (wide loop), copied from Session 15 Slide 18.

SATURATION OF SOFT AND HARD MATERIALS

Slide 18 lists a high B_s for the hard materials, and Slide 19 asks for a high saturation magnetization in the soft ones. The data favour the soft side: on the chart of Slide 19 CoFe and Fe reach about 2 T or more against about 1.5 T for NdFeB, and on Slide 24 soft B_{max} goes up to 2.30 T against 1.23 T for the best hard $\mu_0 M_r$. Both statements are kept as printed; the loops above are the ones of Slide 18.

7.3 Applications [Session 15 Slide 19]

The slide matches each type of loop with its use [Session 15 Slide 19]:

- **Permanent magnets:** hard magnetic materials, with high resistance to demagnetization (high remanence, high coercive field).
- **Materials under alternating electrical fields** (cores of electromagnets, transformers, generators and electric motors, to increase the magnetic flux in coils and circuits): soft magnetic materials, to minimize energy losses, with high saturation magnetization, small coercive field and high permeability.
- **Magnetic memories** in computers, magnetic amplifiers and pulse transformers (data storage): materials with a **rectangular** hysteresis loop, with low remanence, low saturation magnetization and low coercive field (soft magnetic materials modified by a thermal treatment in the presence of a magnetic field).

The slide also plots the saturation magnetization against the coercivity of many materials, from the soft ferrites and iron alloys at low coercivity to the hard ferrites and rare-earth magnets at about 10^5 A/m and above, with the regions labelled soft, semihard and hard. A second figure compares three loops: the soft magnet for electrical applications, the hard magnet for permanent magnets and the square loop for computer applications.

RECTANGULAR LOOP AS PRINTED

Slide 19 prints "low remanance" for the rectangular loop. A rectangular loop has B_r close to B_s, so "low" can only mean low in absolute value, not low relative to B_s. The recording powders of Slide 24 also have H_c of 22 to 240 kA/m, above the 10 kA/m limit of soft materials, so the slide's description does not fit them.

§ 8 Factors that control the hysteresis cycle [Session 15 Slides 20-22]

The shape of the loop depends on [Session 15 Slide 20]:

- **The material's composition:** the number and intensity of its dipoles, and the type of magnetism of the material.
- **Processing:** its microstructure (grain size, pores, impurities, etc.) and the presence of strains due to mechanical deformations.
- **Working conditions:** working temperature, radiation and electrical currents.

The **coercivity** is a strongly microstructure-dependent property (crystallographic direction, grain boundaries, pores, shape of the particles), while the **saturation magnetization** is constant for a material of a given composition (at a given temperature, as the next paragraph shows).

Microstructure [Session 15 Slide 21]. The Bloch walls move more easily in larger grains, and obstacles stop them:

Change	Effect on the Bloch walls	Effect on the loop
Larger grain size	mobility increases	μ increases, H_c and B_r decrease
More inclusions, dislocations, pores, etc.	mobility decreases (a particle pins the wall)	μ decreases, H_c and B_r increase

Temperature [Session 15 Slide 22]. When the temperature rises the mobility of the dipoles rises. The maximum alignment reached when the field is applied is lower, so M_s is reduced, and it is harder to keep the dipoles aligned when the field disappears, so B_r and H_c are reduced (the loop shrinks, as in the slide's figure for low, moderate and high temperature). When $T > T_c$ the dipoles can still align with H, but when H disappears their alignments become random, and the material becomes **paramagnetic**.

§ 9 Curie and Néel temperatures, typical properties [Session 15 Slides 23-24]

Thermal agitation tends to misalign the dipoles [Session 15 Slide 23].

- The **Curie temperature** T_c is the temperature at which the saturation magnetization abruptly falls to 0. For $T > T_c$ ferromagnetic and ferrimagnetic materials become paramagnetic.
- The **Néel temperature** is the temperature at which an antiferromagnetic material becomes paramagnetic: the thermal energy becomes large enough to destroy the macroscopic magnetic ordering of the material.

Ferromagnetic material	Fe	Co	Ni	Gd
T_c in K	1043	1404	631	289

Antiferromagnetic material	α-Mn	MnO	Cr	NiO
Néel temperature in K	100	116	310	523

The slide plots the saturation magnetization of pure iron and of magnetite against the temperature: both fall to zero, iron at about 770 °C (1043 K) and Fe_3O_4 at about 580 °C.

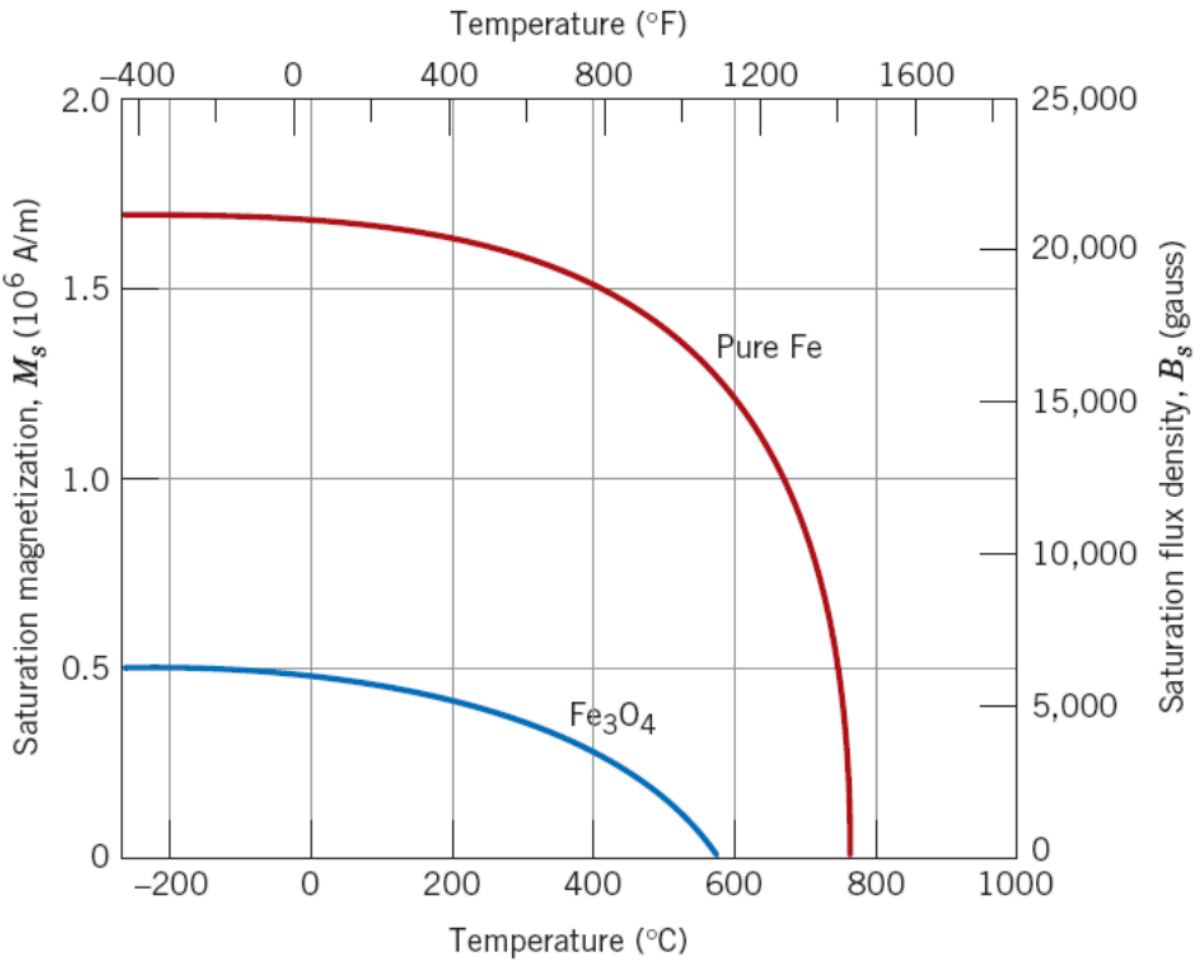


Figure 14. Saturation magnetization and saturation flux density against temperature for pure Fe and Fe3O4, which vanish at the Curie temperature, copied from Session 15 Slide 23.

Slide 24 reproduces three tables of typical properties: selected **soft** magnetic materials (initial and maximum relative permeability, coercivity H_c, retentivity B_r, B_{max} and resistivity: ingot iron, low-carbon steel, Fe-Si alloys, Fe-Ni alloys, Supermendur, metallic glasses, MnZn and NiZn ferrites), selected **hard** (permanent) magnetic materials (Fe-Co, Alnico-5, barium ferrite, $SmCo_5$, $Nd_2Fe_{14}B$, with $\mu_0 M_r$, $\mu_0 H_c$ and $(BH)_{max}$) and typical magnetic recording powders.

TABLE 20-4 ■ *Properties of selected soft magnetic materials*

Name	Composition	Permeability μ_r Initial	Permeability μ_r Maximum	Coercivity H_c(A · m^{-1})	Retentivity B_r (T)	B_{max} (T)	Resistivity ($\mu\Omega$ · m)
Ingot Iron	99.8% Fe	150	5000	80	0.77	2.14	0.10
Low-carbon steel	99.5% Fe	200	4000	100		2.14	1.12
Silicon iron, unoriented	Fe-3% Si	270	8000	60		2.01	0.47
Silicon iron, grain-oriented	Fe-3% Si	1400	50,000	7	1.20	2.01	0.50
4750 alloy	Fe-48% Ni	11,000	80,000	2		1.55	0.48
4-79 permalloy	Fe-4% Mo-79% Ni	40,000	200,000	1		0.80	0.58
Superalloy	Fe-5% Mo-80% Ni	80,000	450,000	0.4		0.78	0.65
2V-Permendur	Fe-2% V-49% Co	800	450,000	0.4		0.78	0.65
Supermendur	Fe-2% V-49% Co		100,000	16	2.00	2.30	0.40
Metglas[a] 2650SC	$Fe_{81}B_{13.5}Si_{3.5}C_2$		300,000	3	1.46	1.61	1.35
Metglas[a] 2650S-2	$Be_{78}B_{13}S_9$		600,000	2	1.35	1.56	1.37
MnZn Ferrite	H5C2[b]	10,000		7	0.09	0.40	1.5×10^5
MnZn Ferrite	H5E[b]	18,000		3	0.12	0.44	5×10^4
NiZn Ferrite	K5[b]	290		80	0.25	0.33	2×10^{12}

TABLE 20-6 ■ *Properties of selected hard, or permanent, or magnetic materials*

Material	Common Name	$\mu_0 M_r$ (T)	$\mu_0 H_c$ (T)	$(BH)_{max}$ (kJ · m^{-3})
Fe-Co	Co-steel	1.07	0.02	6
Fe-Co-Al-Ni	Alnico-5	1.05	0.06	44
$BaFe_{12}O_{19}$	Ferrite	0.42	0.31	34
$SmCo_5$	Sm-Co	0.87	0.80	144
$Nd_2Fe_{14}B$	Nd-Fe-B	1.23	1.21	290–445

TABLE 20-5 ■ *Properties of typical magnetic recording materials in a powder form*

	Particle Length (μm)	Aspect Ratio	Magnetization B_r (Wb/m^2)	Magnetization B_r (emu/cm^3)	Coercivity H_c (kA/m)	Coercivity H_c (Oe)	Surface Area (m^2/g)	Curie Temp. T_c(°C)
γ-Fe_2O_3	0.20	5:1	0.44	350	22–34	420	15–30	600
Co-γ-Fe_2O_3	0.20	6:1	0.48	380	30–75	940	20–35	700
CrO_2	0.20	10:1	0.50	400	30–75	950	18–55	125
Fe	0.15	10:1	1.40[a]	1100[a]	56–176	2200	20–60	770
Barium Ferrite	0.05	0.02 μm	0.40	320	56–240	3000	20–25	350

Figure 15. Properties of selected soft magnetic materials, hard (permanent) magnetic materials and magnetic recording powders, copied from Session 15 Slide 24.

§ 10 Superconductivity and magnetism [Session 15 Slides 25-32]

10.1 Zero resistivity and critical temperature [Session 15 Slide 25]

The electrical resistivity of a **superconductor** becomes **zero** below some critical temperature T_c. Superconductors can be pure metals and alloys, ceramics (typically with copper oxide) and organic compounds (based, for example, on Se or S). T_c ranges between 0.01 and 165 K (the text of the slide; its chart marks 163 K), which is the practical limitation, and it is usually lower for metals than for ceramics. This T_c is the critical temperature of a superconductor, not the Curie temperature of Section 9, and likewise H_c below is the critical field, not the coercivity. The slide gives the critical temperatures (read from the slide):

Material	T_c in K	Material	T_c in K
Al	1.20	Nb_3Sn	18.05
Hg	4.15	Nb_3Al	17.5
Mo	0.92	V_3Ga	16.5
Nb	9.26	La_3In	10.4
Pb	7.19	LaBaCuO	35
Ta	4.48	YBCO	90
Ti	0.39	$Tl_2Ba_2Ca_2Cu_3O_{10}$	125
Zn	0.88	$Hg_{12}Tl_3Ba_{30}Ca_{30}Cu_{45}O_{125}$	138

The slide also plots the resistivity against the temperature (the superconductor drops to zero at T_c, the normal metal tends to a residual value) and the history of the record T_c: the discovery of superconductivity (1911), the heavy-fermion superconductors (1979), the high-T_c cuprates (1986) and the iron-based high-T_c superconductors (2006).

10.2 Critical field and critical current density [Session 15 Slide 26]

At temperatures below T_c the superconducting state stops when a sufficiently large magnetic field (H_c) or current density (J_c) is applied, and H_c and J_c **decrease with increasing temperature**. The **superconducting envelope** is the surface in the (T, H, J) space that bounds the superconducting state; the slide shows it for Nb3Sn, with $B_{c2} = 24.5$ T, $T_c = 18$ K and $J_c = 10^7$ A/cm² ($= 10^{11}$ A/m²).

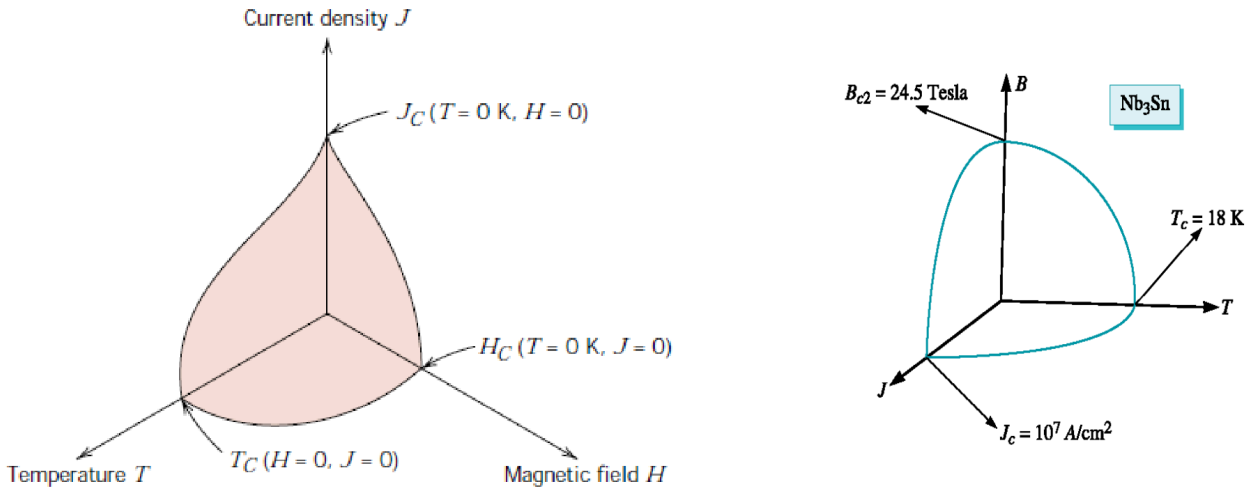


Figure 16. Superconducting envelope: the combined effect of temperature, magnetic field and current density on a Nb3Sn superconductor (Bc2 = 24.5 T, Tc = 18 K, Jc = 10^7 A/cm2), copied from Session 15 Slide 26.

10.3 Meissner effect and levitation [Session 15 Slide 27]

The **Meissner-Ochsenfeld effect** is the expulsion of the magnetic field from a material when it is in the superconducting state. Above T_c the field lines go through the material; below T_c they are pushed around it, and the material becomes diamagnetic. Superconductors in the Meissner state exhibit **perfect diamagnetism** (superdiamagnetism): the total magnetic field is very close to zero deep inside. A permanent magnet **levitates** above a superconductor cooled by liquid nitrogen: the slide's drawing shows the repulsion of the Meissner effect, an attraction from the trapped magnetic flux, and gravity [Session 15 Slide 27].

PRINTED WORDS

Slide 27 writes "semiconductor state" and "Oschsenfeld". The effect belongs to the superconducting state, and the name is Meissner-Ochsenfeld.

10.4 Type I and Type II superconductors [Session 15 Slides 28-29]

- **Type I (soft superconductors):** sharp transition to the normal state, at a very low H_c, so they have low practical interest. They are pure metals: Sn, Al, Pb and Hg [Session 15 Slide 28].
- **Type II (hard superconductors):** gradual transition to the normal state as H grows. The lowest critical field H_{c1} can be 100 times lower than the highest critical field H_{c2}, and they have a high resistance against the destruction of superconductivity by magnetic fields. They are alloys (Nb-Zr, Nb-Ti), intermetallics (Nb_3Sn) and ceramics (YBaCuO). The flux lines begin to penetrate the material at H_{c1}, and at H_{c2} the penetration is complete; for fields between the two the material is in a **mixed state**, with normal and superconducting regions [Session 15 Slide 29].

PRINTED WORDS

Slides 28 and 29 write "Type I semiconductors" and "Type II semiconductors" for the lists of materials. They are superconductors.

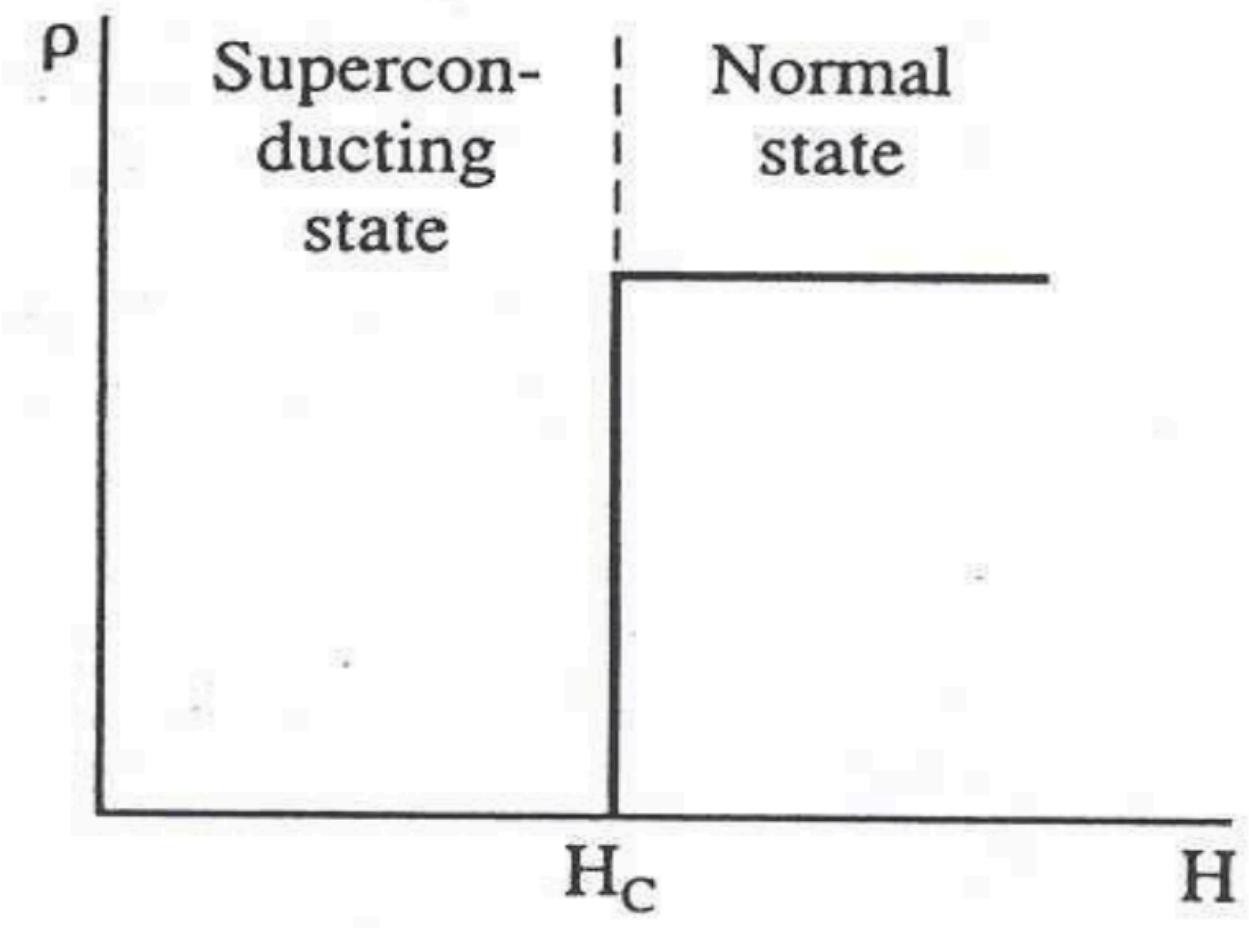


Figure 17. Type I superconductor: resistivity zero in the superconducting state up to the critical field Hc, then an abrupt return to the normal state, copied from Session 15 Slide 28.

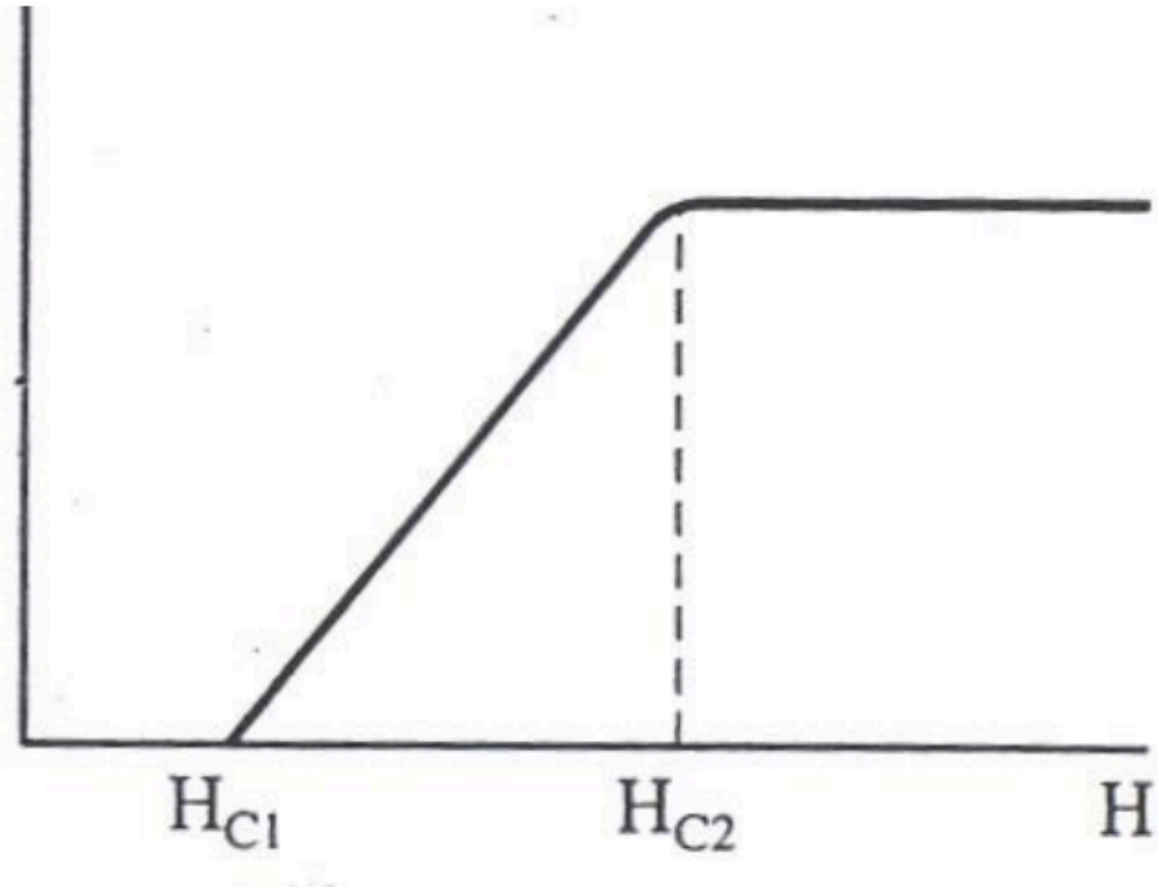


Figure 18. Type II superconductor: gradual transition between the lower critical field Hc1 and the upper critical field Hc2 (the vertical axis has no label on the slide), copied from Session 15 Slide 29.

10.5 Josephson effect and applications [Session 15 Slides 31-32]

The **Josephson** or tunnel effect occurs when a current flows through two superconductors separated by a very thin insulating layer, the **tunnel junction** (an oxide layer of about 3 nm) [Session 15 Slide 31]. The Josephson current goes through the junction even though there is no potential difference; a d.c. voltage applied through the junction causes a radio-frequency current oscillation (and an RF voltage causes a d.c. current); and its value is very sensitive to the magnetic field. It has many applications.

The applications of superconductors on Slide 32 are levitated transport, the powerful superconducting electromagnets of magnetic resonance imaging (MRI), RF and microwave filters (for mobile-phone base stations and for military receivers), the Large Hadron Collider and other particle accelerators, and very low-loss filters based on ceramic superconductors for wireless communications [Session 15 Slide 32].

Videos linked in the slides

These are references printed on the slides, not content of the portal:

- Session 15, Slide 7 (diamagnetism, "The frog incident"): https://www.youtube.com/watch?v=A1vyB-O5i6E&ab_channel=GrahamRumplestiltskinMcLeod
- Session 15, Slide 30 (superconductivity and magnetism): https://www.youtube.com/shorts/Bo8WV032dyk

Part II — Thermal Properties (Session 15, Slides 33–48)

§ 11 Thermal properties: agenda and overview [Session 15 Slide 33]

Slide 33 is the title and the agenda of the thermal part: introduction, heat capacity and specific heat, thermal expansion, thermal conductivity and thermal shock. Its pictures show a thermocouple, a pan (thermal conductivity), a clinical thermometer and the thermal insulation of the space shuttle (thermal insulator).

§ 12 Heat capacity and specific heat [Session 15 Slides 34-35]

12.1 Definitions and phonons [Session 15 Slide 34]

Thermal energy in solids is the increase in the vibrational energy of the atoms, and the vibrations produce travelling lattice waves. A **phonon** is a packet of elastic waves (thermal vibrational energy), with energy

$$E = \frac{hv}{\lambda} = h\omega$$

where h is Planck's constant, v the speed of sound, λ the wavelength and ω the frequency, as printed.

> PRINTED FORM OF THE PHONON ENERGY
> On the slide ω is the ordinary frequency v/λ. With the usual angular frequency $\omega = 2\pi v/\lambda$ the same energy is written $\hbar\omega$, with $\hbar = h/2\pi$ (extension).

The **heat capacity** is the energy required to raise the temperature of **one mole** of a material by one degree (the ability of the material to absorb heat):

$$C = \frac{dQ}{dT}$$

with C in J/(mol·K) (or cal/(mol·K)) and dQ the energy required to produce a temperature change dT. The **specific heat** is the energy required to raise the temperature of **one gram** of a material by one degree; the slide writes it as the heat capacity divided by the atomic weight:

$$c_p = \frac{\text{heat capacity}}{\text{atomic weight}}$$

12.2 Dependence on temperature [Session 15 Slide 35]

The heat capacity **increases with temperature**, because the lattice waves can increase their average energy. Above the **Debye temperature** θ_D the heat capacity becomes independent of temperature, with the approximate value

$$C \approx 3R \approx 6 \text{ cal}/(\text{mol} \cdot \text{K})$$

with R the gas constant. In SI units, $3R = 3 \times 8.314 = 24.9$ J/(mol·K) (the conversion is an extension, with 1 cal = 4.184 J, and it gives 25.1 J/(mol·K) for 6 cal/(mol·K)). The limit of $3R$ per mole is known as the Dulong-Petit value (the name is an extension).

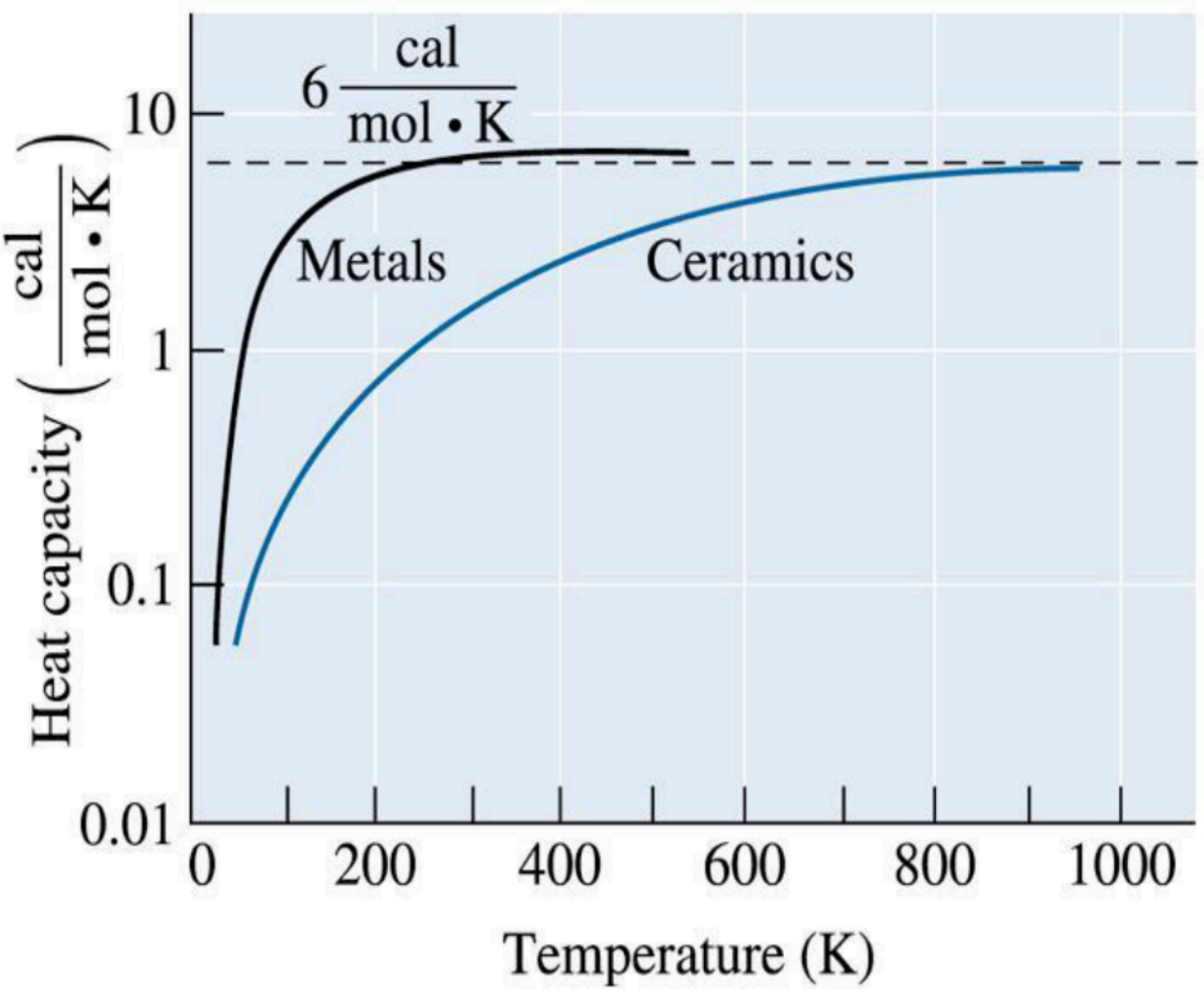


Figure 19. Heat capacity as a function of temperature for metals and ceramics, levelling off near 6 cal/(mol·K) above the Debye temperature, copied from Session 15 Slide 35.

In the chart the metals reach the plateau at a lower temperature than the ceramics.

The heat capacity and the specific heat of most materials **do not depend significantly on the structure** of the material (dislocations, vacancies, grain size). The exception is the **ferromagnetic materials**, such as iron, which show a high specific heat at the Curie temperature (the change from ferromagnetic to paramagnetic) [Session 15 Slide 35].

§ 13 Thermal expansion [Session 15 Slides 36-39]

13.1 Linear coefficient [Session 15 Slide 36]

The **linear coefficient of thermal expansion** α is the amount by which the unit length of a material changes when its temperature changes by one degree:

$$\frac{l_f - l_0}{l_0} = \alpha\,(T_f - T_0) \qquad \text{that is} \qquad \frac{\Delta l}{l_0} = \alpha\,\Delta T$$

α is in °C⁻¹ (or K⁻¹, since only differences of temperature enter). Here $\Delta T = T_f - T_0$. For an isotropic solid (an extension, not printed on the slides), with $\varepsilon = \Delta l / l_0$, $V = l^3$ gives $\Delta V / V_0 = (1+\varepsilon)^3 - 1 = 3\varepsilon + 3\varepsilon^2 + \varepsilon^3 \approx 3\varepsilon$ for small ε, so the volume coefficient is $\alpha_v \approx 3\alpha$.

13.2 Atomic origin [Session 15 Slide 36]

The slide states that materials with a **high bonding energy** (strong atomic bonding) have **low** coefficients of thermal expansion, and plots the interatomic energy against the separation: when the material is heated, the vibration energy rises and the equilibrium spacing between the atoms increases; the increase is smaller for the deep, narrow well of strong bonding than for the shallow well of weak bonding (the same reading as in the note on interatomic potential energy curves).

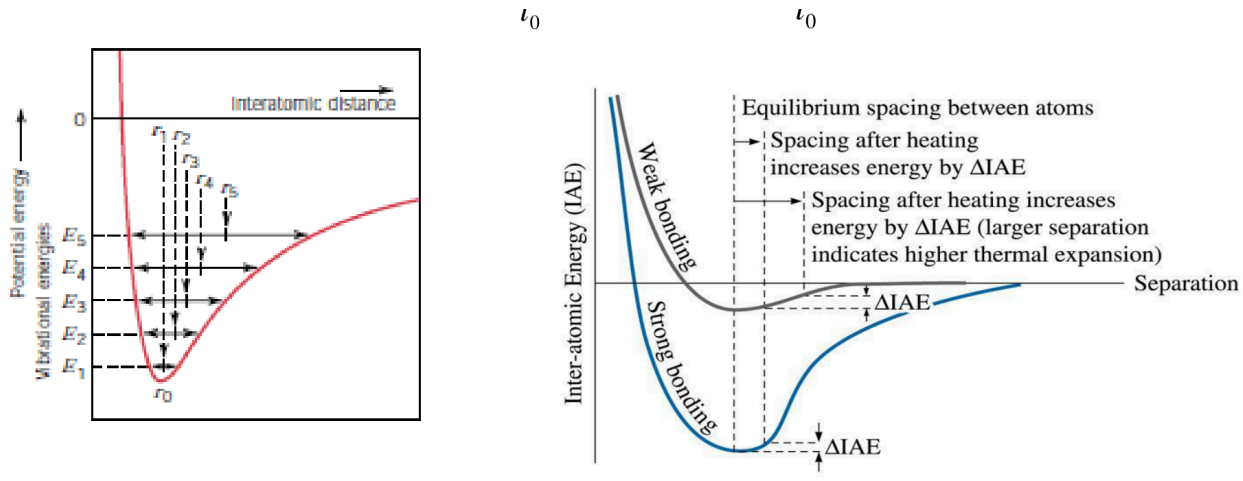


Figure 20. Interatomic energy curves: the spacing grows when the vibration energy grows, and more for weak than for strong bonding; the same slide plots the potential well at several vibrational energies, copied from Session 15 Slide 36.

Why the spacing grows (extension). The reason is the **asymmetry** of the well. Near the minimum the curve is steeper on the compressed side (the repulsion grows fast) than on the stretched side. At a vibration energy E the atom oscillates between a small and a large distance, and the mean of the two lies at a larger

distance than r_0 and moves out as E grows. A symmetric parabola, the harmonic approximation, would keep the mean distance at r_0 and give no thermal expansion at all.

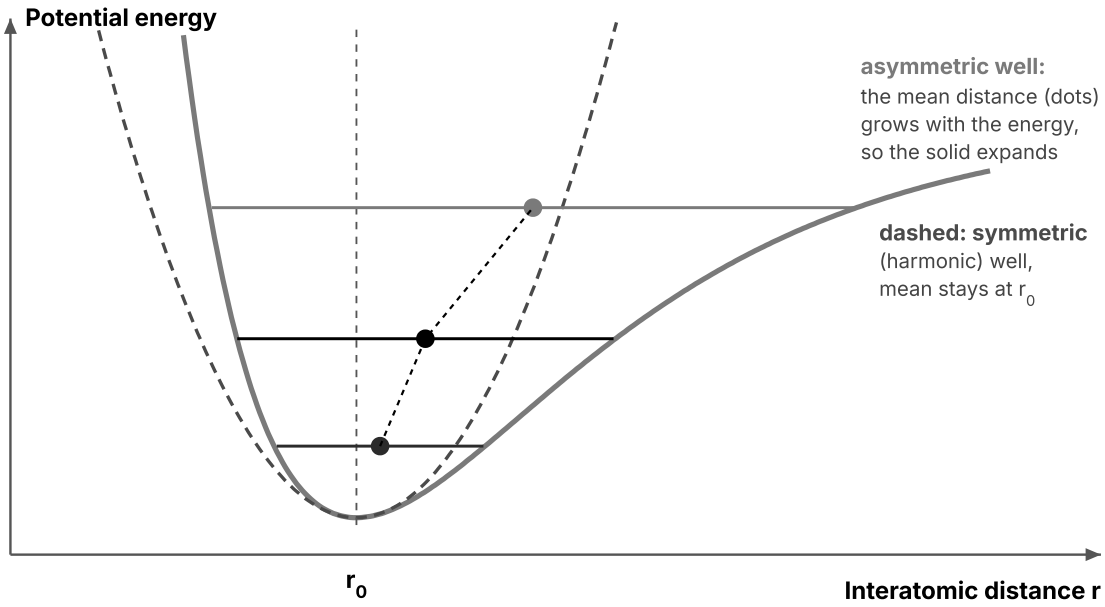


Figure 21. Extension, not printed on the slides: because the real potential well is asymmetric, the mean interatomic distance grows with the vibration energy; a symmetric parabola would keep it at r0 (schematic).

13.3 Values and allotropic changes [Session 15 Slides 37-38]

The slide plots the coefficient of thermal expansion of **iron** against temperature: it changes abruptly at the temperatures where an **allotropic transformation** occurs (the α, γ and δ phases of the figure), and it plots the coefficient of the metals at 25 °C against their **melting temperature**: the higher the melting temperature (the stronger the bonding), the lower the coefficient. Allotropic materials have abrupt changes in their dimensions when phase transformations occur, which cracks refractories [Session 15 Slide 37].

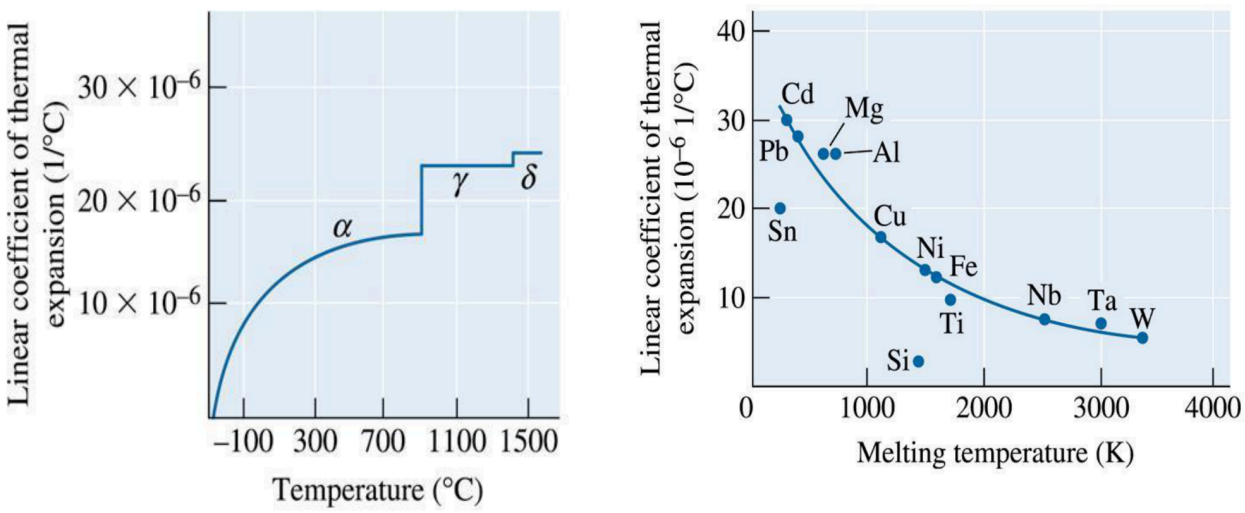


Figure 22. Linear coefficient of thermal expansion of iron against temperature (abrupt changes at the allotropic transformations) and of metals at 25 °C against melting temperature, copied from Session 15 Slide 37.

Slide 38 gives the specific heat c_p, the expansion coefficient α_l and the thermal conductivity k of metals, ceramics and polymers, with the low-expansion alloys **Invar** (64Fe-36Ni, $\alpha_l = 1.6 \times 10^{-6}$ °C⁻¹) and **Super Invar** (63Fe-32Ni-5Co, 0.72×10^{-6} °C⁻¹) boxed. For comparison: aluminium 23.6×10^{-6}, copper 17.0×10^{-6}, iron 11.8×10^{-6}, tungsten 4.5×10^{-6}, alumina 7.6×10^{-6} and fused silica 0.4×10^{-6} °C⁻¹, and 106 to 198×10^{-6} °C⁻¹ for high-density polyethylene.

Material	c_p (J/kg-K)[a]	α_l [(°C)$^{-1}$ × 10^{-6}][b]	k (W/m-K)[c]
		Metals	
Aluminum	900	23.6	247
Copper	386	17.0	398
Gold	128	14.2	315
Iron	448	11.8	80
Nickel	443	13.3	90
Silver	235	19.7	428
Tungsten	138	4.5	178
1025 Steel	486	12.0	51.9
316 Stainless steel	502	16.0	15.9
Brass (70Cu-30Zn)	375	20.0	120
Kovar (54Fe-29Ni-17Co)	460	5.1	17
Invar (64Fe-36Ni)	500	1.6	10
Super Invar (63Fe-32Ni-5Co)	500	0.72	10
		Ceramics	
Alumina (Al_2O_3)	775	7.6	39
Magnesia (MgO)	940	13.5[d]	37.7
Spinel ($MgAl_2O_4$)	790	7.6[d]	15.0[e]
Fused silica (SiO_2)	740	0.4	1.4
Soda–lime glass	840	9.0	1.7
Borosilicate (Pyrex) glass	850	3.3	1.4
		Polymers	
Polyethylene (high density)	1850	106–198	0.46–0.50
Polypropylene	1925	145–180	0.12
Polystyrene	1170	90–150	0.13
Polytetrafluoroethylene (Teflon)	1050	126–216	0.25
Phenol-formaldehyde, phenolic (Bakelite)	1590–1760	122	0.15
Nylon 6,6	1670	144	0.24
Polyisoprene	—	220	0.14

Figure 23. Specific heat cp, linear expansion coefficient and thermal conductivity k of metals, ceramics and polymers (Invar and Super Invar boxed on the slide), copied from Session 15 Slide 38.

13.4 Bimetals [Session 15 Slide 39]

A **bimetal** is made of two metals bonded together with different expansion coefficients. A **bimetallic strip** converts a temperature change into a mechanical displacement: on heating, the metal with the larger coefficient lengthens more and the strip bends towards the other metal (the slide shows copper and iron). Uses: thermostats, thermometers, clocks and electrical devices [Session 15 Slide 39].

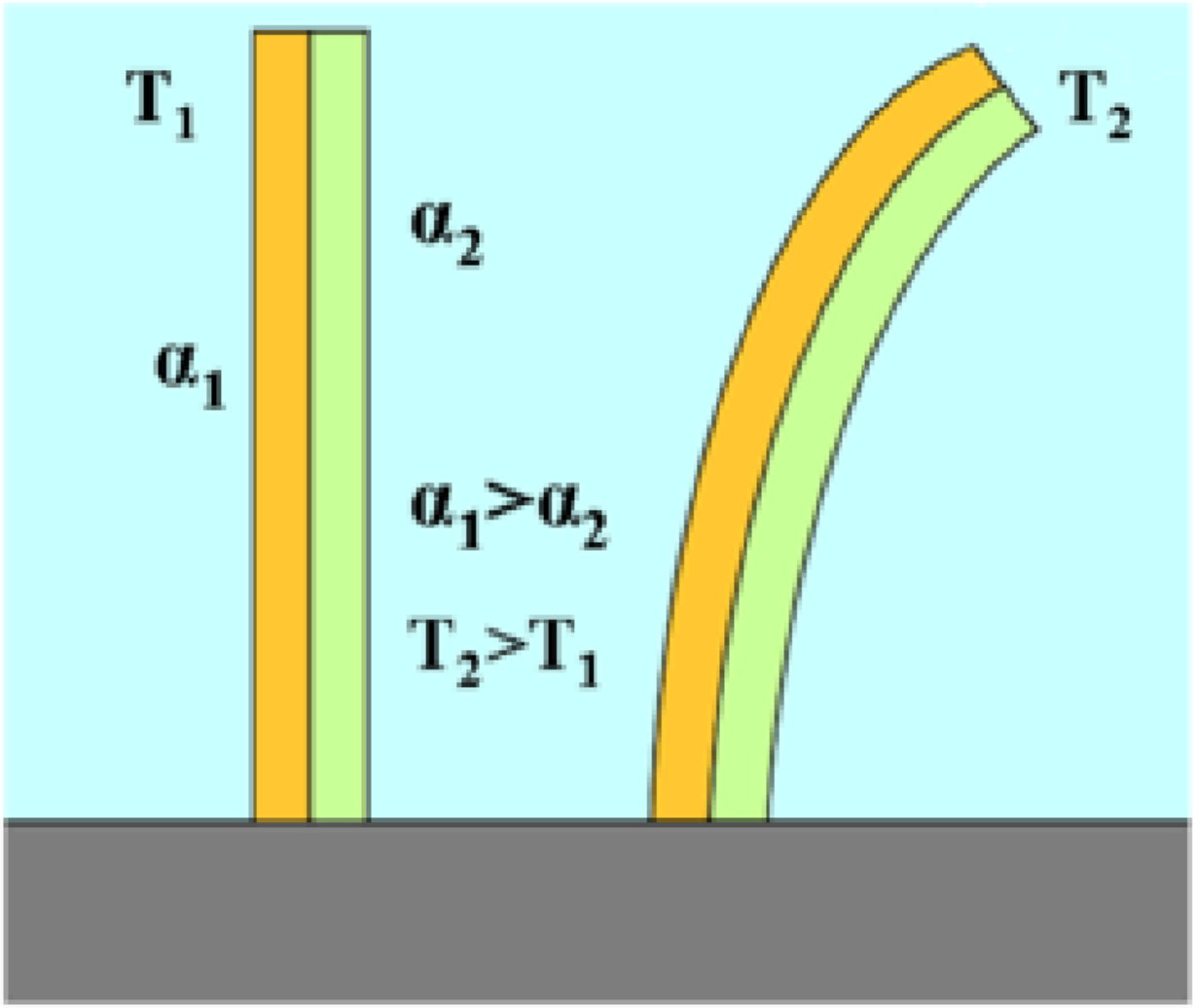


Figure 24. Bimetallic strip of two metals with expansion coefficients α1 > α2: straight at T1 and bent at T2 > T1, copied from Session 15 Slide 39.

§ 14 Thermal conductivity [Session 15 Slides 40-44]

14.1 Fourier's law [Session 15 Slide 40]

The **thermal conductivity** is a microstructure-sensitive property that measures the rate at which heat is transferred through a material. When one end of a bar is heated, a heat flux Q/A flows towards the cold end at a rate set by the temperature gradient in the bar:

$$q = \frac{Q}{A} = -k\,\frac{\Delta T}{\Delta x}$$

q is the heat flux per unit area in W/m², Q the heat flow in W (the slide calls it heat flux), k the thermal conductivity in W/(m·K) and $\Delta T/\Delta x$ the temperature gradient. The minus sign says that heat flows from hot to cold. This is the same structure as the steady-state diffusion law of the note on Fick's first law: a flux

proportional to a gradient, with a transport coefficient (k here, D there); that parallel is an extension.

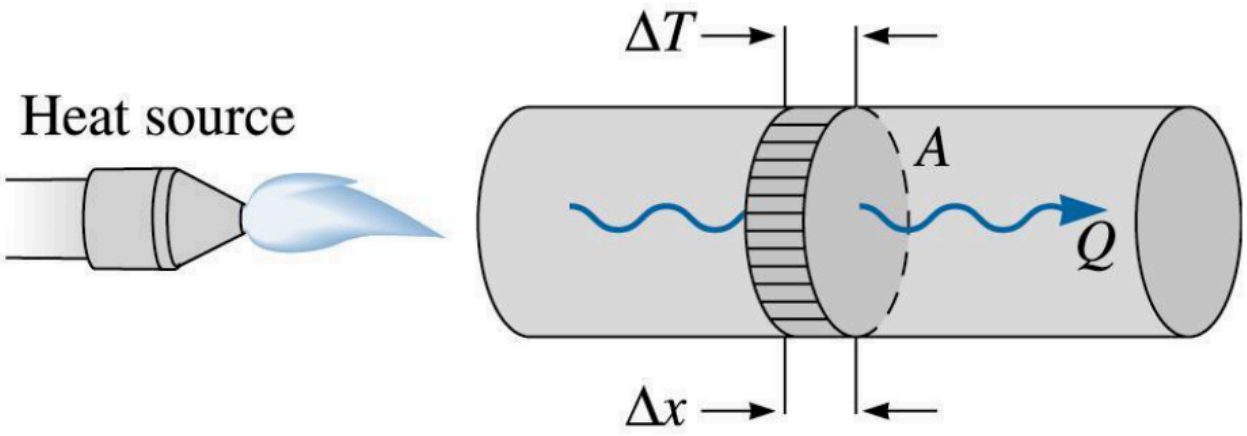


Figure 25. A bar heated at one end by a heat source: the heat flow Q crosses the area A down the temperature difference ΔT over the length Δx, copied from Session 15 Slide 40.

The thermal energy is transferred by two mechanisms: the transfer of **free electrons** (k_e) and the **lattice vibrations or phonons** (k_{ph}):

$$k = k_e + k_{ph}$$

In **conductive materials** the transfer is by free electrons, $k \approx k_e$; in **insulating materials** it is by phonons, $k \approx k_{ph}$.

14.2 Link with the electrical conductivity [Session 15 Slide 41]

The slide gives the relation between the thermal and the electrical conductivity (the **Wiedemann-Franz law**, a name that is an extension):

$$L = \frac{k}{\sigma T}$$

where L is the **Lorenz number** (a constant), $L = 2.44 \times 10^{-8}$ W·Ω/K² (or 5.5×10^{-9} cal·Ω/(s·K²)); the slide calls it the Lorentz number. The units are consistent: $k/(\sigma T)$ is $[\mathrm{W/(m \cdot K)}]/[(\Omega \cdot \mathrm{m})^{-1} \cdot \mathrm{K}] = \mathrm{W \cdot \Omega / K^2}$. The slide also gives a table of the thermal conductivity of pure metals, alloys, ceramics and polymers (1 cal/(cm·s·K) = 418.4 W/(m·K)).

> **DISCREPANCY IN THE PRINTED LORENZ NUMBER**
> The two values of the slide are not the same number: with 1 cal = 4.184 J, 2.44×10^{-8} W·Ω/K² is 5.83×10^{-9} cal·Ω/(s·K²), not 5.5×10^{-9} (a difference of about 6 %). Both are kept as printed; the SI value is the one used on this page.

Material	Thermal Conductivity ($W \cdot m^{-1} \cdot K^{-1}$)	Material	Thermal Conductivity ($W \cdot m^{-1} \cdot K^{-1}$)
Pure Metals:		Ceramics:	
Ag	430		
Al	238	Al_2O_3	16–40
Cu	400	Carbon (diamond)	2000
Fe	79	Carbon (graphite)	335
Mg	100	Fireclay	0.26
Ni	90		
Pb	35	Silicon carbide	up to 270
Si	150	AlN	up to 270
		Si_3N_4	up to 150
Ti	22	Soda-lime glass	0.96–1.7
W	171	Vitreous silica	1.4
Zn	117	Vycor™ glass	12.5
Zr	23	ZrO_2	4.2
Alloys:		Polymers:	
1020 steel	100	6,6-nylon	0.25
3003 aluminum alloy	280	Polyethylene	0.33
304 stainless steel	30	Polyimide	0.21
Cementite	50	Polystyrene	0.13
		Polystyrene foam	0.029
Cu-30% Ni	50	Teflon	0.25
Ferrite	75		
Gray iron	79.5		
Yellow brass	221		

Note: 1 cal/cm · s · K = 418.4 W · m^{-1} · K^{-1}

Figure 26. Thermal conductivity in W/(m·K) of pure metals, alloys, ceramics and polymers, copied from Session 15 Slide 41.

DISCREPANCY BETWEEN THE TABLES

Some values of this table contradict Slides 38 and 43, where alloying lowers k: the 3003 aluminium alloy (280) is above pure Al (238 here, 247 on Slide 38), the 1020 steel (100) is above pure Fe (79) and against 51.9 for the 1025 steel of Slide 38, and yellow brass (221) stands against 120 for 70Cu-30Zn on Slide 38. The values are kept as printed; use them with care.

14.3 Effect of temperature [Session 15 Slide 42]

For the metals, two effects compete. A higher temperature **increases** the contribution of the lattice vibrations k_{ph}, so the thermal conductivity increases; and it **decreases** the electron mobility (more scattering of the electrons), so the thermal conductivity decreases. The chart of the slide shows platinum, aluminium, iron, SiC, Al2O3, glass and a firebrick with porosity.

Two remarks on the chart are extensions. The Wiedemann-Franz law gives $k_e = L\sigma T$, and since σ of a metal falls roughly as $1/T$, k_e stays nearly constant, which fits the flat curve of aluminium. And the phonon conductors SiC and Al2O3 fall with temperature on the chart, so the statement of the slide that k_{ph} grows with T does not hold for every material.

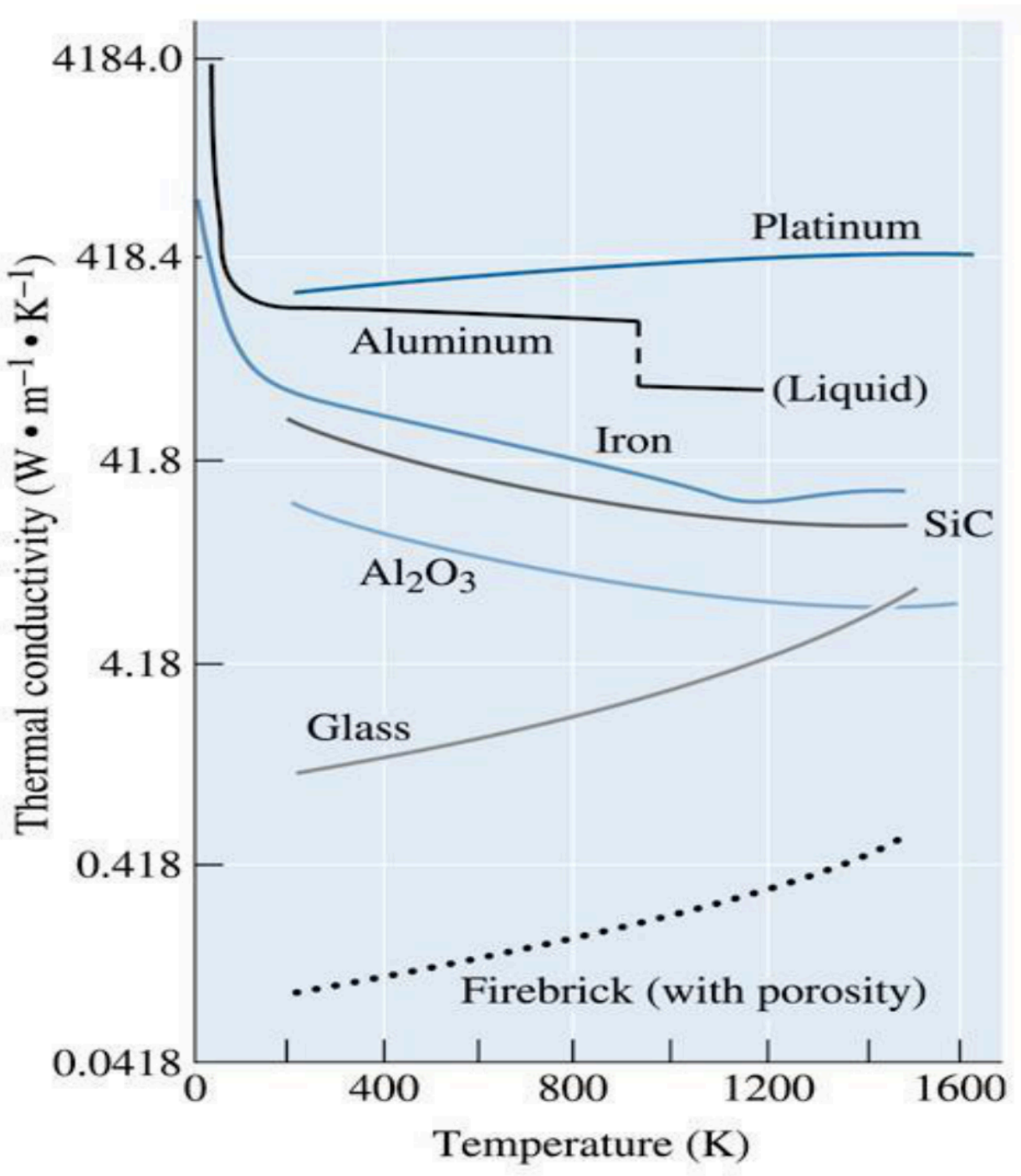


Figure 27. Effect of temperature on the thermal conductivity of several materials (platinum, aluminium, iron, SiC, Al2O3, glass, firebrick), copied from Session 15 Slide 42.

14.4 Effect of defects [Session 15 Slide 43]

Alloying, defects and porosity **reduce** the thermal conductivity, because they scatter electrons and phonons. The slide shows the drop of the thermal conductivity of copper as zinc is added (Cu-Zn), three lattice drawings with impurity atoms and a vacancy, and the pores of a ceramic (micrograph of the porosity).

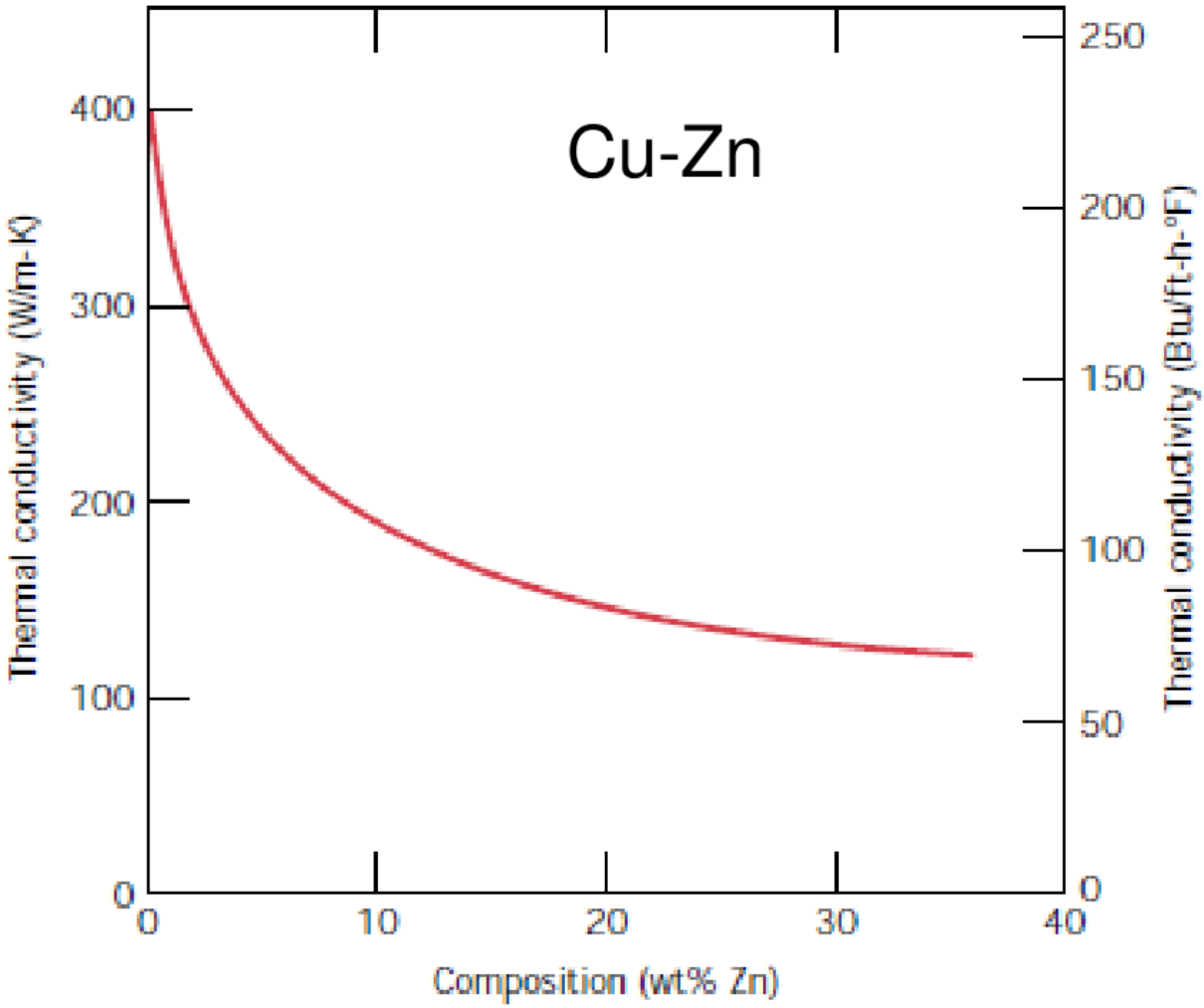


Figure 28. Thermal conductivity of Cu-Zn alloys against zinc content: alloying lowers the thermal conductivity, copied from Session 15 Slide 43.

14.5 Thermocouples [Session 15 Slide 44]

A **thermocouple** is a temperature-measuring device made of two dissimilar conductors that contact each other at one or more spots. It produces a voltage when the temperature of one of the spots differs from the reference temperature of the other parts of the circuit. The slide shows a junction at 300 °C that gives 12.2 mV.

PRINTED LEGS

The slide labels the two conductors Ni-Cr and Al-Cr. The second label is kept as printed; the usual pair for this combination is Ni-Cr against Ni-Al (extension).

§ 15 Thermal stresses in constrained bodies [Session 15 Slides 45-46]

Thermal stresses are the stresses introduced into a material by differences in the amount of expansion or contraction that occur because of a temperature change. In **confined materials** they result from restrained thermal expansion or contraction [Session 15 Slide 45]:

$$\sigma = E\,\alpha\,(T_0 - T_f) = E\,\alpha\,\Delta T$$

where E is Young's modulus (MPa), α the linear coefficient of thermal expansion (°C⁻¹) and σ the thermal stress (MPa). The slide calls ΔT the thermal gradient; in this expression it stands for $T_0 - T_f$, the initial minus the final temperature, which is the opposite sign of the $\Delta T = T_f - T_0$ of Section 13. The slide states the signs:

- **Heating**, $T_f > T_0$: the stress is **compressive** ($\sigma < 0$).
- **Cooling**, $T_f < T_0$: the stress is **tensile** ($\sigma > 0$).

The origin is the free expansion that the walls prevent (this link with Hooke's law is an extension). A free bar would have the thermal strain $\alpha(T_f - T_0)$. Held at constant length, the total strain is zero, $0 = \sigma/E + \alpha(T_f - T_0)$, and therefore $\sigma = E\alpha(T_0 - T_f)$. Dimensionally $[E\alpha\Delta T] = \text{MPa} \cdot {}^\circ\text{C}^{-1} \cdot {}^\circ\text{C} = \text{MPa}$.

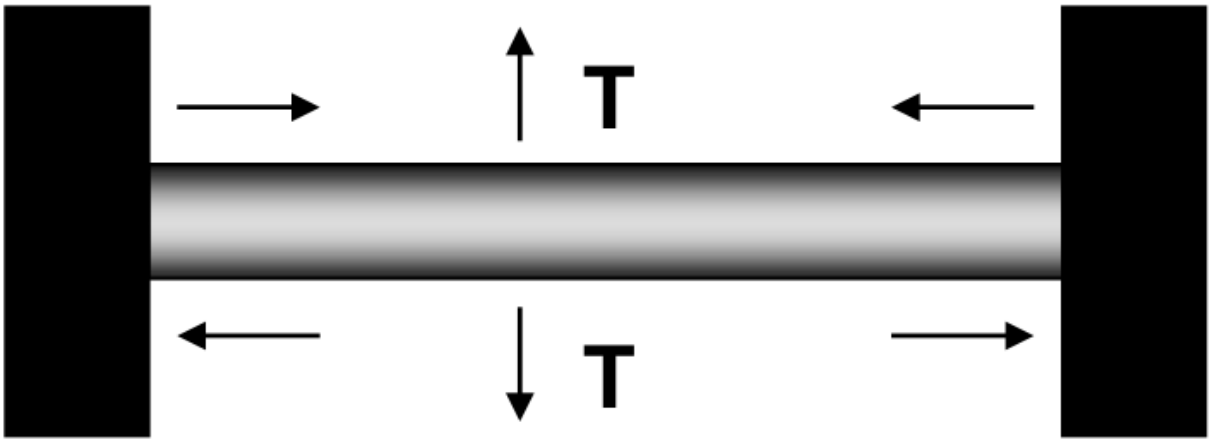


Figure 29. Bar confined between two rigid walls, so that its thermal expansion or contraction under a temperature change T is restrained, copied from Session 15 Slide 45.

Slide 46 ("Problem 5. Topic Mechanical properties") applies the law to problem 5 of the Topic 5 sheet. A steel wire, initially 1 m long, is stretched by a force of 1400 N at 20 °C. After the force is applied the wire is held at constant length and heated to 40 °C, and the diameter is asked for that makes the thermal expansion stress compensate the applied stress. With the data of the sheet ($E = 207$ GPa, $\alpha_l = 12.5 \times 10^{-6}$ °C⁻¹) the magnitude of the thermal stress is $E\alpha_l|T_f - T_0| = 51.75$ MPa (compressive, because the wire is heated), the area is $A = F/\sigma = 1400\ \text{N}/51.75\ \text{N/mm}^2 = 27.053\ \text{mm}^2$ and the diameter is $d = \sqrt{4A/\pi} = 5.87$ mm (5.869 mm; the official key prints 5.86 mm). The full step-by-step solution, with the check against the official key, is in Problem T5-05.

SLIDE 46

On the rendered slide the box under the statement is empty, so the data of the sheet are used here. The PDF does keep a solution in its hidden text layer, with slips: it prints $\alpha = 12.5 \times 10^{-3}$ °C⁻¹ although its 51.75 MPa needs 12.5×10^{-6}, it writes the area in mm, and it rounds r to 2.93 mm before doubling it, which is where the 5.86 mm comes from.

§ 16 Thermal shock [Session 15 Slides 47-48]

16.1 Temperature gradients [Session 15 Slide 47]

Thermal stresses also appear without any external constraint when a body has a **temperature gradient**, caused by rapid heating or cooling, because the outside temperature changes more rapidly than the interior [Session 15 Slide 47]:

- **Rapid heating:** the exterior is hotter and expands, so it is under **compressive** surface stresses and **tensile** interior stresses.
- **Rapid cooling:** the exterior contracts, so it is under **tensile** surface stresses and **compressive** interior stresses.

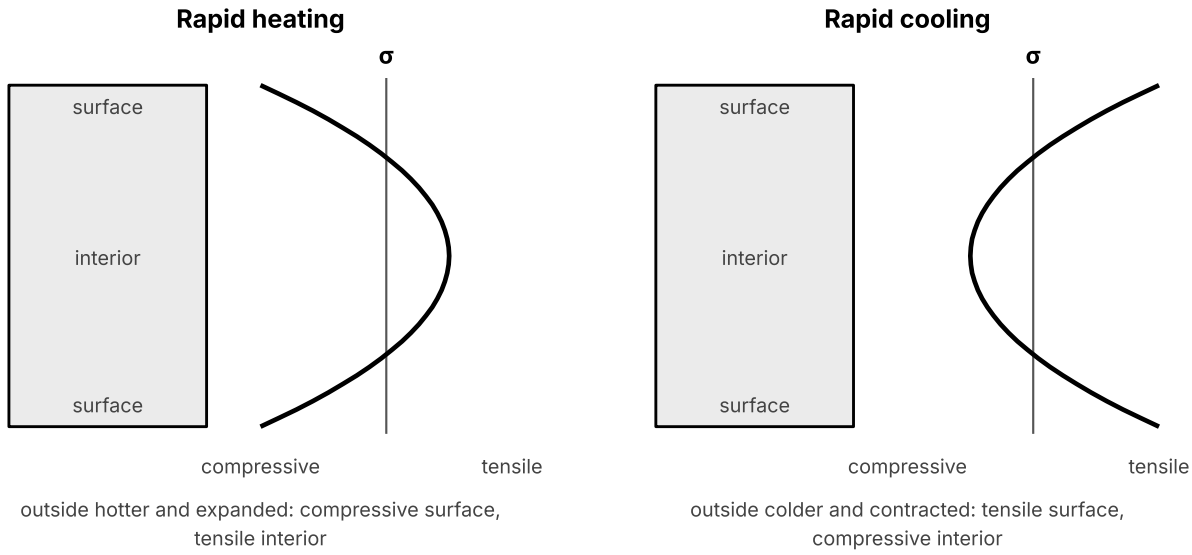


Figure 30. Rapid heating makes the surface compressive and the interior tensile; rapid cooling reverses the signs. The stress profile is an extension, not printed on Slide 47 (schematic).

16.2 Thermal shock resistance [Session 15 Slide 48]

Thermal shock is the failure of a material caused by stresses introduced by sudden changes in temperature, and it is typical of **brittle materials**. The thermal shock resistance of the slide is

$$TSR = \frac{\sigma_f \, k}{E \, \alpha_l}$$

with σ_f the fracture stress, k the thermal conductivity, E the modulus of elasticity and α_l the linear coefficient of thermal expansion [Session 15 Slide 48]. The factors that raise the resistance, as listed on the slide:

Factor	Change	Effect on the thermal shock resistance
Coefficient of thermal expansion α_l	lower	higher
Thermal conductivity k	higher	higher
Modulus of elasticity E	lower	higher
Fracture stress σ_f	higher	higher
Phase transformations	dimensional changes	thermal shock is worse

The formula reproduces these four trends: a low α_l and a low E make the thermal stress $E\alpha_l\Delta T$ small, and a high k evens out the temperature so the gradient is small, while a high σ_f raises the stress the material can stand. The units of $\sigma_f k/(E\alpha_l)$ are W/m (the unit check is an extension).

§ 17 Collected formulas of the topic

Everything below is taken from the sections above; it is a quick reference, not new content.

Quantity	Formula	Where
Induction and permeability	$B = \mu H = \mu_0 H + \mu_0 M,\ \mu_r = \mu/\mu_0$	Section 3
Susceptibility	$M = \chi_m H,\ \mu_r = 1 + \chi_m$	Section 3
Bohr magneton	$\mu_B = 9.27 \times 10^{-24}$ A·m²	Section 2
Hysteresis limit between soft and hard	$H_c = 10\,000$ A/m	Section 7
Heat capacity	$C = dQ/dT,\ C \approx 3R$ above θ_D	Section 12
Thermal expansion	$\Delta l/l_0 = \alpha\,\Delta T$, with $\Delta T = T_f - T_0$	Section 13
Fourier's law	$q = Q/A = -k\,\Delta T/\Delta x,\ k = k_e + k_{ph}$	Section 14
Wiedemann-Franz law (name: extension)	$L = k/(\sigma T),\ L = 2.44 \times 10^{-8}$ W·Ω/K²	Section 14
Thermal stress	$\sigma = E\alpha(T_0 - T_f) = E\alpha\Delta T$, with $\Delta T = T_0 - T_f$	Section 15
Thermal shock resistance	$TSR = \sigma_f k/(E\alpha_l)$	Section 16